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To cite this article: J. S. Yogesh Kumar, Prabhakar Rai & Arya Sen (21 Sep 2026): Polymorphism in *Umbonium vestiarium* (Linnaeus, 1758) from the Sundarban Biosphere Reserve: morphological, morphometric and distributional analyses, Molluscan Research, DOI: [10.1080/13235818.2026.2735667](https://doi.org/10.1080/13235818.2026.2735667)

To link to this article: <https://doi.org/10.1080/13235818.2026.2735667>



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Polymorphism in *Umbonium vestiarium* (Linnaeus, 1758) from the Sundarban Biosphere Reserve: morphological, morphometric and distributional analyses

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ABSTRACT

The intertidal gastropod *Umbonium vestiarium* (Linnaeus, 1758) shows remarkable shell colour variation throughout the Indo-Pacific region, but no detailed quantitative study had previously examined this polymorphism in the ecologically diverse Sundarban Biosphere Reserve (SBR), West Bengal, India. This study provides the first comprehensive assessment of shell colour polymorphism in the SBR using multiple analytical approaches across six intertidal islands representing different environmental conditions of the Indian Sundarbans. A total of 4033 individuals were collected from 60 quadrats ($25 \times 25 \text{ cm} = 1/16 \text{ sq. metre}$), and eleven distinct colour morphs (P1–P11) were identified through a standardised 25-character binary matrix, which resolved three colour-pattern-based morph groups by UPGMA cluster analysis. Morphometric analysis of 330 specimens revealed significant shape variation among morphs, and discriminant analysis separated all eleven colour-defined morphs in shape space. However, clustering based on shape alone did not match colour-based groupings, showing shell shape and colour vary along decoupled axes. Uneven morph distribution across islands suggests environmental influence, consistent with environmentally structured divergence in the Sundarbans, although without proof. Morphs P4 and P9 are potential conservation priorities, and this frequency data offers a baseline for future biodiversity monitoring.

ARTICLE HISTORY

Received 20 May 2026
Final version received 15
September 2026

KEYWORDS

Shell polymorphism;
protected area; malacology;
morphometrics; mosaic
selection hypothesis;
intertidal biodiversity

Introduction

Phenotypic polymorphism is the co-occurrence of two or more distinct, discontinuous phenotypic variants within a single interbreeding population, representing one of the most compelling and enduring problems in evolutionary biology (Leimar 2005). In most cases, both genetic and environmental factors interact to generate and maintain such variation, with natural selection, genetic drift and phenotypic plasticity each contributing to the equilibrium of morph frequencies observed in nature (Leimar 2009). Snail shells are surprisingly powerful tools for science. Their durable, patterned exteriors make them perfect for studying why so much natural variety exists within a single species. The grove snail *Cepaea nemoralis* has been a go-to example for decades, revealing how predators, geography, and genetics shape shell colours and bands (Jones *et al.* 1977; Saenko *et al.* 2021). In marine environments, the intertidal snail *Umbonium vestiarium* (Linnaeus, 1758) serve a similar role boasting one of the most spectacular ranges of shell colour variation seen across Indo-Pacific molluscs (Grüneberg 1980; Chan *et al.* 2016).

Umbonium vestiarium, commonly known as the button top snail or button shell, is a small to

medium-sized trochid gastropod belonging to the family Trochidae (Order Trochida, Class Gastropoda, Phylum Mollusca). The species is distributed across the Indo-West Pacific region from the Mediterranean Sea and the Red Sea in the west, where it has been documented as an introduced species following Lessepsian migration (Barash & Danin 1972), along the coasts of Arabia, the Indian subcontinent and South-east Asia (Kalayanasundaram *et al.* 1974), to Japan in the east (Tamaki & Kikuchi 1983). In the Indian context, it is recognised as one of the dominant macrobenthic invertebrates of clean, well-sorted, medium-grained sandy intertidal beaches, frequently accounting for 34–56% of total macrobenthic abundance and biomass at stations where it occurs (Sivadas *et al.* 2012). The species is a filter and deposit feeder (Fretter 1975) inhabiting the upper few centimetres of sand at the low- to mid-intertidal zone, where it is subject to intensive predation by naticid, olivid gastropods and asteroid sea stars (Berry 1984). The species plays an important economic role, being harvested for local consumption and collected for curio manufacture (Poutiers 1998; Appukuttan & Ramadoss 2000), and its shell has been proposed as a novel bio-adsorbent for metal ion removal from aqueous solutions (Hossain & Maxwell 2022).

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Supplemental data for this article can be accessed online at <https://doi.org/10.1080/13235818.2026.2735667>.

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The remarkable shell colour diversity of *U. vestiarium* first came under serious scientific scrutiny when Grüneberg (1980) examined around 38,000 individuals across India, Sri Lanka, Malaysia and Hong Kong. He identified two broad types of variation: clear-cut, classifiable patterns driven by major genes, and a more fluid 'pseudo-polymorphism' where environment and polygenic background blur the lines between morphs. He catalogued at least 14 named forms such as, 'candy stripe', 'pearl', 'bronze' and showed that morph variety differed noticeably between locations (Grüneberg 1980). Subsequent investigations across the species' Indo-Pacific range have corroborated and substantially extended this picture. Sivadas *et al.* (2012), working on intertidal populations at Kalbadevi Bay, Ratnagiri (west coast of India), documented 11 colour pattern morphs, of which three occurred in low abundance. They have found that morph frequencies were broadly consistent between the northern and southern ends of the study beach, with the exception of one morph that was disproportionately represented at the southern stations, an observation consistent with fine-scale spatial variation in selection pressure or substrate composition (Sivadas *et al.* 2012). Along the Bangladesh coast, Hossain and Maxwell (2022) identified four broad colour morphs, each showing considerable internal variation, consistent with the pseudo-polymorphic nature of the species first described by Grüneberg (1980). Across all studied populations, morph counts range widely, from just four to 11 (Sivadas *et al.* 2012), up to 24 in southeastern India, and a remarkable 96 in Singapore (Chan *et al.* 2016) highlighting how strongly location shapes visible diversity. That Singapore figure is particularly impressive, surpassing morph counts recorded in *Littorina saxatilis*, *Cepaea nemoralis* and *Tridacna crocea*. Chan *et al.* (2016) also showed that the two Singapore populations shared statistically similar morph frequencies, pointing to free interbreeding and little differential selection between nearby sites.

The mechanisms responsible for maintaining such remarkable intraspecific polymorphism in *U. vestiarium* remain incompletely resolved. The classic explanation of polymorphism is, predators favour common morphs, protecting rarer ones is unlikely here, since its main predators (naticid and olivid gastropods, and sea stars) hunt using chemoreception rather than sight, making visual-based selection improbable (Chan *et al.* 2016; Berry 1984). Thermoregulatory selection has been proposed as an alternative explanatory framework; in sun-baked tropical shores, darker shells overheat more easily, giving paler morphs a survival edge (Miura *et al.* 2007; Chan *et al.* 2016). Developmental plasticity and pseudo-polymorphism, in which a single genotype can generate multiple phenotypes through epigenetic responses

to environmental conditions such as substrate chemistry, temperature fluctuations, or ontogenetic growth interruptions, also appears to contribute significantly to the overall phenotypic diversity of the species (Grüneberg 1980). Additionally, substrate heterogeneity, micro-habitat partitioning and background colour matching are likely to generate spatially varying selection pressures that maintain different morphs at different levels within complex intertidal mosaics (Noda *et al.* 1995; Tamaki & Kikuchi 1983). The relative contributions of these mechanisms are expected to vary among populations and habitats, underscoring the importance of site-specific, integrative investigations.

The Sundarban Biosphere Reserve (SBR), nestled at the mouth of the Ganga–Brahmaputra–Meghna delta on the Bay of Bengal, is one of the world's most ecologically complex intertidal landscapes. Spanning roughly 9630 km², including vast mangrove forests, tidal creeks, mudflats and sandy beaches; it holds UNESCO World Heritage and Ramsar Wetland status, and supports extraordinary marine biodiversity (Hait & Behling 2019; Habib *et al.* 2021). Despite this ecological significance, the taxonomic and ecological documentation of its invertebrate macrobenthos remains incomplete, and morphological diversity within intertidal molluscan species has not been systematically characterised (Habib *et al.* 2021). The pronounced environmental gradients within the SBR, ranging from high-salinity, wave-exposed outer islands with clean sandy beaches to turbid, brackish inner islands fringed by mangrove roots and mudflats (Sahana *et al.* 2022; Samanta *et al.* 2021) create a richly heterogeneous selection landscape that is theoretically favourable for the maintenance of high intraspecific phenotypic diversity in intertidal inhabitants such as *U. vestiarium*. With ongoing changes including rising temperatures, increasing salinity and shifting mangrove composition (Sahana *et al.* 2016; Samanta *et al.* 2021), establishing baseline data on shell colour polymorphism here is increasingly urgent; morph frequencies could serve as early indicators of ecosystem stress. Yet despite *U. vestiarium*'s well-documented colour diversity across the Indo-Pacific, no study has ever examined it within a mangrove-estuarine system like the Sundarbans, nor combined colour classification with shell shape analysis and spatial distribution (Grüneberg 1980; Chan *et al.* 2016; Hossain & Maxwell 2022). This study addresses that gap directly.

This study pursued four clear goals: (i) taxonomic description of the species from the region and catalogue all observed colour morphs using a standardised system based on band and whorl patterns; (ii) measure how similar or different those morphs are from one another, using clustering and ordination techniques; (iii) test whether colour morphs also differ in physical shell shape by analysing six shell

measurements across morphs; (iv) examine how all 11 morphs are distributed across six islands in Sundarban. Taken together, these four approaches offer an unusually integrated view of *U. vestiarium* polymorphism in Sundarban.

Materials and methods

Study area and site selection

The present study was conducted within the Sundarban Biosphere Reserve (SBR), located at the southern apex of the Ganga–Brahmaputra–Meghna delta system along the Bay of Bengal coast, West Bengal, India (21°31′–22°30′N, 88°05′–89°05′E).

Six islands (Sagar Island, Jamboo Island, Bakkhali, G Plot, Lothian Island and Dhanchi Island) within the SBR were selected as study sites on the basis of their contrasting habitat characteristics, accessibility during sampling surveys and the confirmed presence of established *U. vestiarium* populations (Kalayanasundaram *et al.* 1974; Sivadas *et al.* 2012). Summary characteristics of each sampling site are presented in Table 1 and Figure 1.

Field sampling and specimen collection

Sampling was carried out during low tide across all six study islands, with 10 randomly placed 0.25 m² quadrats per island within the intertidal sandy zone, following established protocols (Tamaki & Kikuchi 1983; Sivadas *et al.* 2012). All *U. vestiarium* individuals visible on or within the upper 2–3 cm of sediment was hand-collected, as the species rarely burrows deeper (Sivadas *et al.* 2012). Specimens were labelled and transported on ice for laboratory processing. Across 60 quadrats in total (10 replicates × 6 islands), 4033 individuals were collected for subsequent analyses.

Specimen Preparation; Identification; Morph Recognition; Character Scoring; Measurements;

Once specimens reached the laboratory, live shells were relaxed in a 7% magnesium chloride solution for around 30 min to allow soft tissue removal without shell damage. Shells were then cleaned in a 10% hydrochloric acid solution for 2–3 min to remove sediment, organic matter and any surface encrustations that might obscure colour patterns,

before being rinsed in distilled water and air-dried. Any shells with mechanical damage covering more than 20% of the patterned surface were discarded and replaced, ensuring a minimum of 29 specimens per morph for morphometric work. From the remaining intact pool, 30 adult specimens per morph were drawn blindly from the sample containers without visual inspection for size, height, or pattern prominence. While this haphazard sampling procedure was designed to prevent conscious investigator bias towards larger or more aesthetic shells, formal mathematical randomisation (such as random-number generation across catalogued individuals) was not performed. All cleaned shells were photographed under standardised daylight-balanced lighting at a fixed focal distance, with a millimetre scale bar included in every image.

All specimens were confirmed as *U. vestiarium* based on a suite of diagnostic shell features including a smooth, highly polished surface, 5–6 whorls, a pearlescent white interior aperture, and most distinctively, the calcareous umbilical callus, a polished white disc sealing the umbilicus that serves as the primary diagnostic character for the genus (Grüneberg 1980; Hossain & Maxwell 2022). Identification followed Linnaeus (1758), Grüneberg (1980) and Chan *et al.* (2016).

Individual colour morphs were classified using a standardised binary character matrix adapted from Grüneberg (1980), Sivadas *et al.* (2012) and Chan *et al.* (2016). Each specimen was independently assessed by two observers under calibrated daylight conditions (5500 K LED), with a third assessment used in ambiguous cases (under 3% of specimens). This yielded 11 stable, reproducible morphs (P1–P11) with over 97% interrater agreement.

Morphological variation across morphs was captured using 25 binary characters, scored as present (1) or absent (0) and organised into Band Design (9 characters) and Whorl Design (16 characters) complexes following Chan *et al.* (2016). This presence/absence matrix treats each morph as a discrete phenotypic unit, consistent with standard numerical taxonomy practice for categorical data (Sneath & Sokal 1973).

For each of the 11 morphs, 30 specimens were selected for morphometric measurement, yielding a total sample of $n = 330$ specimens. All measurements

Table 1. Summary of the six study islands within the Sundarban Biosphere Reserve, West Bengal, India.

Island	Coordinates (approx.)	Habitat Type	Tidal Exposure	Quadrats (n)	Total N
Sagar Island	21°38′N, 88°04′E	Exposed sandy intertidal flat	Meso-macro tidal	10 × 11 morphs	656
Jamboo Island	21°51′N, 88°13′E	Mixed sand-mudflat, mangrove edge	Meso-tidal	10 × 11 morphs	534
Bakkhali	21°34′N, 88°12′E	Broad sandy beach, moderate wave energy	Macro-tidal	10 × 11 morphs	635
G Plot	21°44′N, 88°21′E	Turbid estuarine-sandy flat	Meso-tidal	10 × 11 morphs	547
Lothian Island	21°42′N, 88°23′E	Stable outer island, fine sandy beach	Meso-macro tidal	10 × 11 morphs	826
Dhanchi Island	21°37′N, 88°25′E	Outer island, low-energy sandy shore	Meso-tidal	10 × 11 morphs	835



Figure 1. Map of the study site.

were made using a calibrated digital Vernier caliper with a precision of 0.01 mm, following measurement protocols described by Marcus (1990), Bookstein (1991) and Rohlf and Marcus (1993) for gastropod linear morphometrics. Four primary raw shell parameters were recorded for each specimen: Shell Diameter (SD), Shell Height (SH), Aperture Width (AW) and Spire Height (SpH) (Table 2). Following the recommendations of Baur and Leuenberger (2011), derived ratios and percentage indices were excluded from multivariate inferential analyses to prevent artificial weight inflation and avoid multicollinearity among morphometric predictors..

Selected representative samples were deposited in the National Zoological Collection of the Zoological Survey of India, Sunderban Regional Center, with the following registration numbers: KN2619, KN4216, KN4253, KN4703, KN4704, KN4805, KN5058.

Data analysis

Morphological similarity and classification

To quantify differences between the 11 colour morphs based on their binary character scores, pairwise morphological similarity was calculated using the Jaccard similarity coefficient (Jaccard 1901), defined as $J(A, B) = a / (a + b + c)$, where a represents characters shared by both morphs, b characters exclusive to morph A, and c characters exclusive to morph B. The

Jaccard coefficient was preferred to the Simple Matching Coefficient (SMC) because it excludes shared character absences from the similarity calculation a critical

Table 2. Shell morphometric parameters measured in *Umbonium vestiarium* ($n = 330$ specimens across 11 morphs).

Parameter	Abbreviation	Definition and Measurement Protocol	Units
Shell Diameter	SD	Maximum linear diameter measured at the equatorial plane of the body whorl, perpendicular to the axis of coiling.	mm
Shell Height	SH	Axial length from apex to base of the shell, measured parallel to the coiling axis.	mm
Aperture Width	AW	Maximum internal width of the aperture, measured at its widest point perpendicular to the columellar axis.	mm
Spire Height	SpH	Distance from the apex to the suture of the last whorl, along the coiling axis.	mm
AW as % of SD	AW%SD	Proportional aperture width index: $(AW/SD) \times 100$. Reflects relative aperture broadness.	% (unitless)
SpH as % of SH	SpH%SH	Proportional spire height index: $(SpH/SH) \times 100$. Reflects relative spire development.	% (unitless)
Elongation Index	EI	Derived ratio SH/SD . Values approaching 0 indicate planispiral (flat) shells; values approaching 1 indicate turbanate (tall) shells.	ratio

advantage when working with sparse binary presence/absence matrices where many characters are absent in both morphs for trivial rather than biologically meaningful reasons (Sneath & Sokal 1973). The SMC was nonetheless computed alongside the Jaccard matrix for comparative transparency with other studies. Jaccard dissimilarity ($D = 1 - J$) was derived for all subsequent clustering operations, and mean pairwise similarity per morph was computed as an index of each morph's overall phenotypic isolation from the rest.

Hierarchical cluster analysis was performed on the Jaccard dissimilarity matrix using the Unweighted Pair Group Method with Arithmetic Mean (UPGMA) algorithm (Sneath & Sokal 1973). Cluster solutions at $k=3$ and $k=4$ were assessed, and dendrogram fidelity was evaluated using the cophenetic correlation coefficient. All clustering computations were implemented via the `scipy.cluster.hierarchy` module in Python 3 (Virtanen *et al.* 2020). Cluster number (k) was selected using three complementary lines of evidence rather than a single automated criterion: (i) the cophenetic correlation coefficient, assessing overall dendrogram fidelity; (ii) the average silhouette width computed on the Jaccard distance matrix for $k=2-6$; and (iii) the fusion-height gap statistic, identifying the k -solution(s) that remain stable over the widest range of dissimilarity thresholds before the next fusion event. Fine-grained solutions ($k=4-5$) achieved marginally higher silhouette and plateau-width scores than $k=3$, but produced no additional character-based synapomorphies beyond those already diagnostic of the three-group solution (Table 4) and further fragmented the numerically dominant Group I into subsets of one or two morphs, precluding meaningful group-level statistical comparison in the downstream morphometric and distributional analyses. We therefore adopted $k=3$ as the coarsest solution that (a) is statistically well-supported (cophenetic $r=0.83$; silhouette $=0.24$, comparable to finer partitions), (b) is anchored to independently interpretable synapomorphies (e.g., the translucent pattern unique to Group II; pink banding unique to Group III), and (c) retains group sizes ($n \geq 2$ morphs) sufficient for the group-level analyses that follow.

Principal Coordinates Analysis (PCoA; classical multidimensional scaling) was applied directly to the Jaccard dissimilarity matrix. Eigenvalues were extracted from the double-centred squared dissimilarity matrix, and scores on the first two principal coordinate axes (PCo1 and PCo2) were retained for graphical display. PCoA was implemented using `sklearn.manifold` and `scipy` routines in Python 3.

To identify which of the 25 binary characters carried the most diagnostic information for distinguishing morphs, Shannon entropy (H) was computed for each character based on its frequency of presence p across the 11 morphs: $H = -p \log_2 p - (1-p) \log_2 (1-p)$.

Entropy values range from 0 (character universally present or absent; maximally informative) to 1.0 (character present in exactly half of morphs; maximally variable and therefore uninformative for discrimination). Following established thresholds in numerical phenetics (Sneath & Sokal 1973), characters with $H < 0.70$ were designated as diagnostic. Within this category, characters with frequency ≤ 0.20 were classified as rare autapomorphies (restricted to one or two morphs), while those with frequency ≥ 0.80 were classified as conserved synapomorphies shared across nearly all morphs.

Shell morphometric analysis

Prior to formal statistical testing, descriptive statistics including mean, standard deviation, standard error, minimum, maximum and coefficient of variation were computed for each of the six shell parameters across all 11 morphs. Normality within each morph-parameter group was assessed using the Shapiro-Wilk test (Shapiro & Wilk 1965), and homogeneity of variances was evaluated using Levene's test (Levene 1960), with non-parametric alternatives adopted where significant heteroscedasticity was detected.

Inter-morph differences across each raw morphometric parameter were tested using one-way Analysis of Variance (ANOVA; $df = 10, 319$) for normally distributed, homoscedastic data, with the Kruskal-Wallis test applied as a confirmatory non-parametric analysis for all parameters regardless of distributional properties. Effect sizes were quantified using eta-squared (η^2), interpreted following Cohen (1988): $\eta^2 < 0.06 = \text{small}$, $0.06-0.14 = \text{medium}$, $> 0.14 = \text{large}$. All tests were conducted in SciPy (Virtanen *et al.* 2020) at a significance threshold of $\alpha = 0.05$. To visualise overall morphometric variation across morphs without collinearity artefacts, Principal Component Analysis (PCA) was applied to a standardised (z-scored) matrix of 330 specimens $\times$ 4 raw linear parameters (SD, SH, AW, SpH). Components with eigenvalues exceeding 1.0 (Kaiser criterion) were retained, and component loadings were examined to identify the morphometric variables most responsible for inter-morph differentiation (Pedregosa *et al.* 2011). Linear Discriminant Analysis (LDA) was subsequently performed on the same standardised 4-variable data, using morph identity (P1–P11) as the class variable, to determine the degree to which raw shell dimensions alone correctly classified specimens into their assigned morphs. Canonical scores were plotted in LD1–LD2 space with 95% confidence ellipses calculated for each individual morph (P1–P11) as well as for the centroids of the three UPGMA colour-pattern groups (Group I–III) to visualise separation and overlap in shape space (Baur & Leuenberger 2011).

To compare morphometric groupings with those derived from colour-pattern clustering, Ward's

linkage hierarchical clustering (Ward 1963) was applied to a matrix of standardised mean morphometric profiles across the 11 morphs. Pairwise Pearson correlations were additionally computed among all six morphometric parameters to identify growth-module couplings, and allometric relationships between shell height and diameter were examined by Ordinary Least Squares regression, with slopes compared against the isometric expectation of 1.0 (Gould 1966; Marcus 1990).

Distributional analysis

Spatial distribution of morphs across the six islands was assessed using Pearson chi-square contingency analysis applied to the 11×6 observed count matrix ($df = 50$), with effect size estimated using Cramér's V . Standardised residuals were computed per cell to identify morph-island combinations having significant chi-square contributions. Per-morph inter-island abundance variation was further evaluated using one-way ANOVA ($df = 5, 54$) and confirmed with the Kruskal-Wallis test. Morph diversity at each island was characterised using the Shannon-Wiener index (Shannon & Weaver 1949), Simpson's diversity index (Simpson 1949), and Pielou's evenness index (Pielou 1966). Spearman rank correlation between total island abundance and Shannon H' was used to test whether larger populations supported higher morph diversity.

Integration and software

Findings across all four analytical frameworks; taxonomic, morphological, morphometric and distributional were synthesised through a composite guild assignment per morph, integrating convergent evidence from Jaccard UPGMA clustering, Ward morphometric clustering and distributional abundance data. All analyses were performed in Python 3, using NumPy, pandas, SciPy (Virtanen *et al.* 2020), scikit-learn (Pedregosa *et al.* 2011), matplotlib and seaborn. Selected analyses were independently verified using PAST 4.03 (Hammer *et al.* 2001), with preliminary data management conducted in Microsoft Excel (version 365).

Results

Shell description and morph characterisation

General shell morphology

Specimens of *U. vestiarius* collected from the intertidal sandy zones of the Sundarban Biosphere Reserve conform to the diagnostic characters of the species, presenting a small to medium-sized, depressed-conical shell; wider than tall; with a height mean of 3.7 mm and a mean shell diameter of 10.8 mm (Figure 2). The spire is low to moderately elevated,

with an apical angle of $90-120^\circ$, and the apex is smooth and obtuse. Whorls number 3–6, their surfaces sculptured only by fine incremental growth lines, with sutures shallow but well-defined. The body whorl is large, accounting for approximately 70–85% of total shell height, and is convex above and flattened to slightly concave below. The most diagnostic external feature of the genus is the umbilical callus; a conspicuous, polished, white to cream-coloured calcareous disc that completely occludes the umbilicus on the basal surface compared to other species of the same genus; which have granulated, or strongly ribbed sculpturing of umbilical callus. In all examined specimens, this callus was circular in outline, smooth and glossy and demarcated from the surrounding basal shell surface by a subtle colour contrast. The aperture was obliquely ovate, pearly white internally, with a thin and simple outer lip. The columella was smooth and slightly arched, merging imperceptibly with the calloused umbilical region. The operculum was corneous, multispiral, thin and light brown, with a central nucleus and fine concentric growth lines. The teleoconch surface was uniformly smooth and highly polished, with a vitreous to porcelain-like lustre; a character diagnostic for this species and the primary substrate, where the species' varied chromatic morphism is expressed. No axial ribs, spiral cords, or nodular sculpture were present. Growth striae were fine, visible only under magnification, and run parallel to the outer lip. In larger individuals, slight peripheral angulation of the body whorl may be noted, though the junction between the dorsal and basal surfaces remains broadly rounded without a distinct keel.

Identification and Characterisation of eleven polymorphic forms

Examination of sampled individuals revealed eleven phenotypically distinct colour morphs (P1–P11), each representing a unique combination of band design and whorl design characters. Morph characterisation was based on nine discrete morphological characters organised into two functional character complexes: (A) Band pattern, comprising ground colour, ornamental line type, stripe position and band position; and (B) Whorl pattern, comprising apical surface colour, basal surface colour, bottom edge character, overall pattern expressivity and whorl-side pattern expressivity (Figure 3). The complete character matrix for all 11 morphs is presented in Table 3, and photographic representations of each morph are shown in Figure 2 (P1–P11).

Band colouration was dichotomous: nine morphs exhibit brown as the ground colour (P1–P3, P5–P9, P11), while two (P4, P10) bore a pink background. Ornamental line types fall into two categories: wavy-line design (P1, P3, P5, P9, P10, P11) or uniform stripes (P2, P4, P6, P7, P8). In the great majority of



Figure 2. 11 morphs of *Umbonium vestiarium* recorded from six islands of Sundarban Biosphere Reserve. P1-P11: Number of morphs; t: top view; b: bottom view; s: side view; a: aperture. (Scale 1 mm).

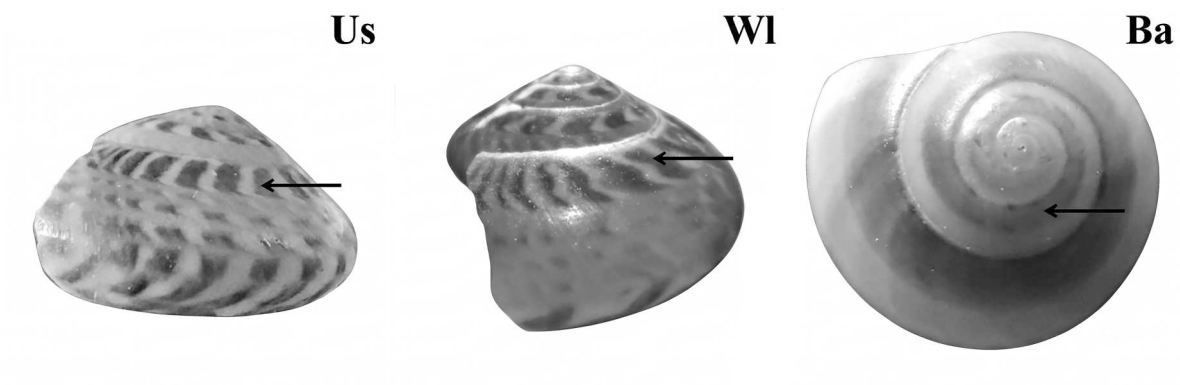


Figure 3. Prominent characters observed in *Umbonium vestiarium*. Us: Uniform stripes; Wl: Wavy lines; Ba: Band.

morphs (10 out of 11), the principal pigment band is positioned at the top of the whorl. The single exception P8 bears its band at the bottom of the whorl, a morphological novelty that makes this morph unambiguously distinct from all others. Pattern expressivity is prominent in six morphs (P2, P3, P5, P6, P8, P10), translucent in two (P4, P7), and mixed in three (P1, P9, P11). Translucent morphs display substantially reduced overall pigment density, with the shell surface appearing semi-glassy, and are exclusively confined to Group II in the subsequent cluster analysis.

Whorl surface colouration exhibits considerable diversity. Four distinct colours were recorded on the apical surface (brown, pink, grey, white), and four on the basal surface (brown, pink, grey, white). Among the most ecologically distinctive colour states, the exclusively grey-on-grey surface of P11 is unique within the sampled population and is associated with the highest Shannon entropy value in the diagnostic character analysis. The white-on-white P6 represents another distinctive pallid phenotype, occurring at highest frequency on the outer marine-influenced islands.

Table 3. Morphological character matrix for eleven colour morphs of *Umbonium vestiarium* from the Sundarban Biosphere Reserve.

Morph	Band Colour	Band Type	Stripe Position	Band Position	Top Colour	Bottom Colour	Bottom Edge	Overall Pattern	Side Pattern
P1	Brown	Wavy lines	Absent	Top	Brown	Brown	Solid	Mixed	Mixed
P2	Brown	Uniform stripes	Top of whorl	Top	Pink	Brown	Patterned	Prominent	Prominent
P3	Brown	Wavy lines	Top of whorl	Top	Grey	White	Patterned	Prominent	Translucent
P4	Pink	Uniform stripes	Absent	Top	Grey	Brown	Solid	Translucent	Translucent
P5	Brown	Wavy lines	Bottom of whorl	Top	White	Pink	Solid	Prominent	Prominent
P6	Brown	Uniform stripes	Bottom of whorl	Top	White	White	Patterned	Prominent	Prominent
P7	Brown	Uniform stripes	Absent	Top	Pink	Pink	Solid	Translucent	Translucent
P8	Brown	Uniform stripes	Top of whorl	Bottom	Pink	Brown	Patterned	Prominent	Prominent
P9	Brown	Wavy lines	Bottom of whorl	Top	Brown	Brown	Patterned	Mixed	Mixed
P10	Pink	Wavy lines	Bottom of whorl	Top	Pink	White	Solid	Prominent	Prominent
P11	Brown	Wavy lines	Bottom of whorl	Top	Grey	Grey	Solid	Mixed	Mixed

Presence/absence of nine morphological characters across morphs P1–P11. Band types: wavy lines (undulating pigment bands), uniform stripes (parallel linear stripes). Pattern expressivity scored as prominent (dense pigmentation), translucent (reduced pigmentation), or mixed (intermediate).

Morphological polymorphism analysis

Pairwise Jaccard similarity

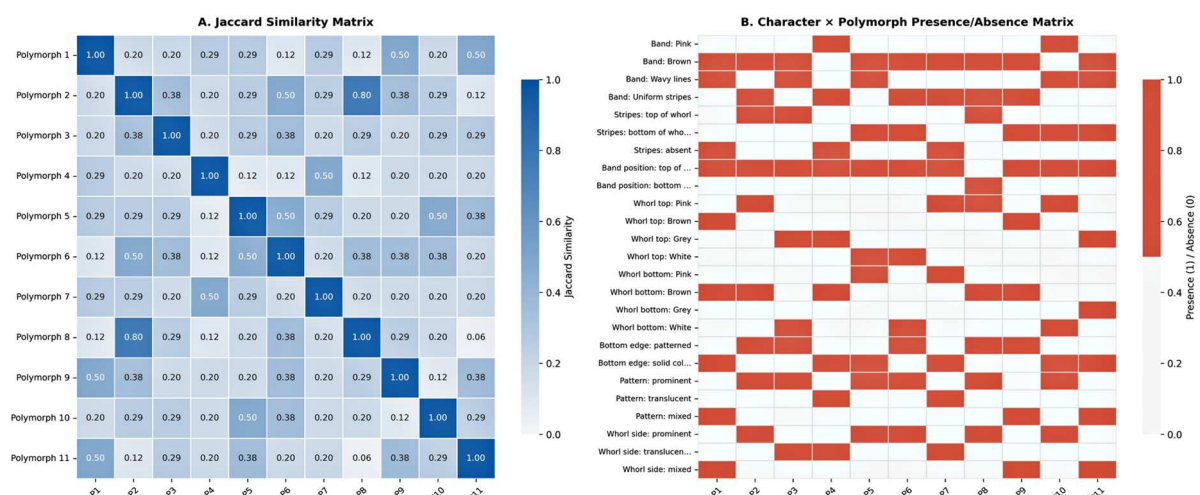
Pairwise morphological similarity among the 11 morphs, assessed using the Jaccard similarity coefficient (J) on 25 characters, revealed substantial inter-morph divergence (Figure 4). Jaccard values ranged from $J = 0.059$ (P8–P11; most dissimilar pair) to $J = 0.800$ (P2–P8; most similar pair), with an overall mean pairwise similarity of 0.272 ± 0.113 (SD). The highest pairwise similarity is attributable to shared brown banding, uniform stripe type and prominent pattern expression; a character suite shared by the two most abundant morphs in the population. The most dissimilar pair, P8 and P11, are virtually complementary in character state distribution: P8's basal-band position, patterned bottom edge, and brown ground colour contrast sharply with P11's apical-band, solid edge, grey-on-

grey surface and mixed pattern expression. Morph 4 recorded the lowest average Jaccard similarity of all morphs (mean $J = 0.216$), identifying it as the most morphologically isolated taxon in the dataset; a finding subsequently corroborated by PCoA ordination.

UPGMA cluster analysis and morph groups

UPGMA hierarchical clustering applied to the Jaccard dissimilarity matrix ($D = 1 - J$) resolved three primary morph groups at a dissimilarity threshold of approximately 0.60 (Figure 5). As discussed in hierarchical clustering of mean morphometrics, the three-group topology recovered here from colour-pattern characters is not consistently reproduced by independent clustering of shell morphometrics.

Group I (P2, P3, P5, P6, P8, P10; $n = 6$) is the largest and most phenotypically diverse assemblage, united

**Figure 4.** A: Jaccard similarity matrix heatmap. B: Character × morph presence/absence heatmap. Blue = present; light = absent.

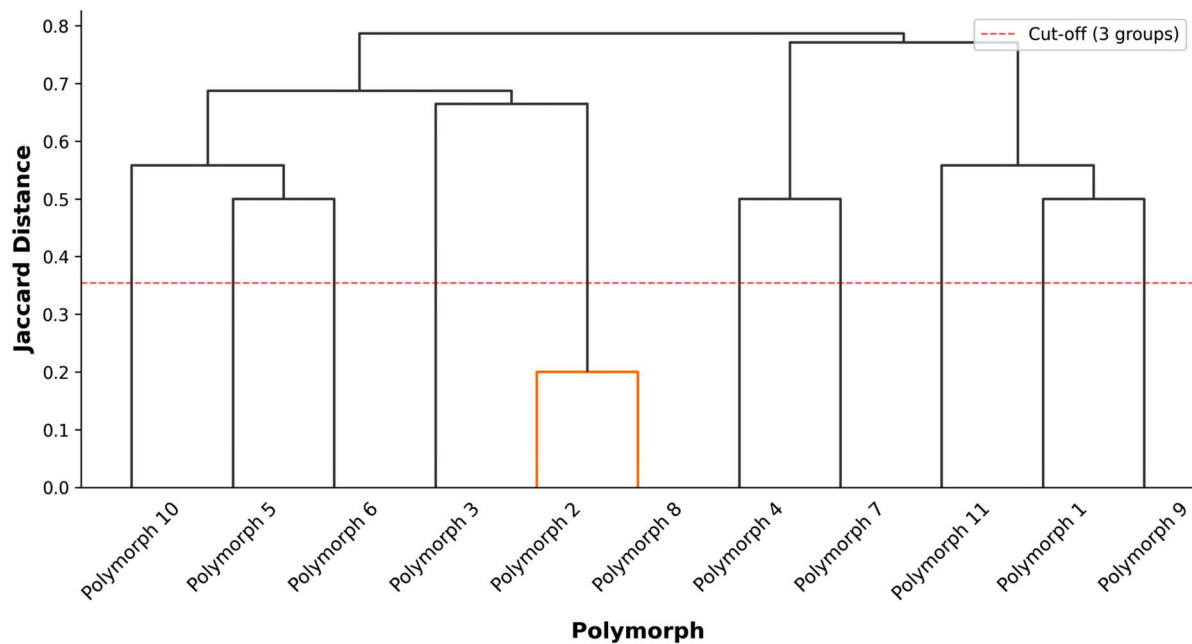


Figure 5. UPGMA dendrogram of 11 *Umbonium vestiarium* morphs based on Jaccard dissimilarity. Red dashed line = cut-off for 3-group solution.

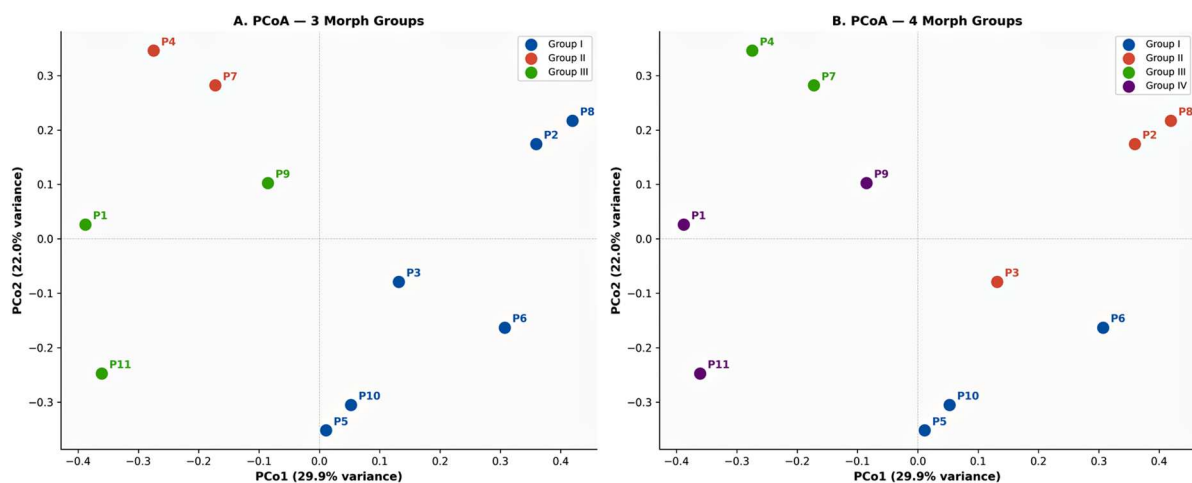


Figure 6. PCoA ordination of 11 *Umbonium vestiarium* morphs. A: 3-group solution. B: 4-group solution. Variance explained: PCo1 = 29.9%, PCo2 = 22.0%.

by brown banding and the conserved character of a band at the top of the whorl. This group accounts for 49.9% of total individual abundance across the six sampled islands and is ecologically associated with heterogeneous high-intertidal sandy flats. A four-cluster solution subdivides Group I into sub-clusters: group Ia (P2, P3, P8), differentiated by stripes at the top of the whorl and a patterned bottom edge, and group Ib (P5, P6, P10), distinguished by ventral stripe position and variable edge patterning.

Group II (P4, P7; $n = 2$) is the smallest and most morphologically cohesive unit, uniquely defined by the co-occurrence of uniform stripes and translucent pattern expression; this is a combination absent from all other morphs. This group contributes 23.3% of total

abundance, with P7 acting as the dominant wide-spread morph and P4 as a micro-habitat specialist.

Group III (P1, P9, P11; $n = 3$) is characterised by pink band elements, wavy-line ornamentation, and mixed whorl-side patterns; and is morphometrically characterised by higher elongation indices (descriptive morphometrics), contributing 26.8% of total abundance with a preference for sheltered intertidal micro-environments.

Principal coordinates analysis

PCoA ordination (Figure 6) on the Jaccard dissimilarity matrix resolved the three morph groups in two-dimensional phenotypic space. PCo1 (29.9% of total variance) broadly separates Group II morphs (positive scores)

from Groups I and III (negative to neutral scores), reflecting the unique uniform-stripe/translucent character profile of P4 and P7. PCo2 (22.0% of variance) separates Group III (positive scores) from Group I (negative scores), corresponding to the pink-versus-brown ground colour axis.

Morph 4 occupies the most extreme position in ordination space, confirming its status as the most morphologically isolated morph; consistent with its lowest average Jaccard similarity (0.216) and its unique combination of pink ground colour, uniform stripes, grey apical surface and translucent pattern. P2 and P8 cluster in close proximity, consistent with their highest pairwise similarity ($J = 0.800$). The four-group PCoA panel further resolves the sub-structure within Group I, with the southern island specialists P5, P6 and P10 forming a tight sub-cluster and P2, P3, P8 forming another, separated primarily along the aperture width and stripe-position axes.

Diagnostic character analysis

Shannon entropy analysis ($H < 0.70$) identified nine diagnostic characters among the 25 scored (Table 4). The most conserved character was Band position: top of whorl (frequency = 0.91, $H = 0.44$), present in all morphs except P8, qualifying it as the strongest shared diagnostic character, for *U. vestiarium* in the Sundarban population. Its complementary character, Band position: bottom of whorl (frequency = 0.09, $H = 0.44$), is a unique character state exclusive to P8 and represents the single most morphologically distinctive character state recorded in this study. Similarly, Whorl bottom: Grey (frequency = 0.09, $H = 0.44$) is a unique character restricted to P11, consistent with the unique grey-on-grey surface colouration of that morph.

Pattern: translucent and Whorl side pattern: translucent are co-restricted to Group II (P4, P7), functioning as synapomorphies for that clade. Band: Brown (frequency = 0.82, $H = 0.68$) is the most conserved

pigmentation character overall, a most widespread state shared across Groups I and III. In total, 16 of 25 characters showed intermediate entropy ($H \geq 0.70$) and are classified as variable, reflecting the high phenotypic plasticity of this species across the Sundarban tidal mosaic.

Shell morphometric analysis

Descriptive morphometrics

Quantitative shell morphometry was conducted on 330 specimens ($n = 30$ per morph) across six parameters: shell diameter (SD), shell height (SH), aperture width (AW), spire height (SpH), aperture width as percentage of shell diameter (AW%SD), and spire height as percentage of shell height (SpH%SH). The elongation index ($EI = SH/SD$) was also computed per morph as a summary shape descriptor (P1–P11: 0.44, 0.211, 0.449, 0.238, 0.307, 0.174, 0.352, 0.166, 0.494, 0.514, 0.613). Descriptive statistics are presented in Figure 7.

Shell diameter ranged from a morph mean of 8.64 ± 0.86 mm in P1– 13.57 ± 1.31 mm in P7; a 56% difference in body size across morphs. Shell height was least in P6 (2.32 ± 0.16 mm; $EI = 0.174$), the most compressed morph, and greatest in P11 (5.78 ± 0.38 mm; $EI = 0.613$), the tallest morph. This near-four-fold difference in EI between P6 and P11 documents a morphological gradient from nearly planispiral to markedly turbinate shell architecture within a single species. Aperture width was correspondingly lowest in P1 (1.51 ± 0.12 mm) and highest in P11 (6.19 ± 0.35 mm), again a near-fourfold range. The AW%SD index was most extreme in P11 (68.1%), nearly double the Group I average (30.4–38.7%), highlighting P11's unusually wide aperture relative to its body size.

Group-level comparisons reveal coherent morphometric identities. Group I morphs (P2, P3, P5, P6, P8, P10) consistently show the largest mean shell diameter (10.71 mm) with a relatively compressed shell height (mean $EI = 0.279$), producing a broad, depressed

Table 4. Diagnostic characters identified by Shannon entropy analysis ($H < 0.70$) from 25 scored binary morphological characters.

Character	Frequency	Shannon H	Classification	Taxonomic / Ecological Significance
Band position: top of whorl	0.91	0.44	Shared diagnostic character	Near-universally conserved; present in all morphs except P8; most informative species-level character
Band position: bottom of whorl	0.09	0.44	Unique character state	Unique to P8; represents a derived, morphologically novel condition
Whorl bottom: Grey	0.09	0.44	Unique character state	Unique to P11; associated with distinct pigmentation pathway
Band: Brown	0.82	0.68	Widespread character state	Dominant, highly conserved ground colour across all morph groups
Band: Pink	0.18	0.68	Shared diagnostic character (Grp III)	Rare; restricted to P1, P9, P11; likely a distinct carotenoid-based pigmentation
Whorl top: Brown	0.18	0.68	Diagnostic character	Rare; linked to Group I morphs (P1, P9)
Whorl top: White	0.18	0.68	Diagnostic character	Restricted to P5, P6; associated with low-insolation or low-salinity substrates
Whorl bottom: Pink	0.18	0.68	Diagnostic character	Restricted to P5, P7; character shared across groups
Pattern: Translucent	0.18	0.68	Shared diagnostic character (Grp II)	Exclusive to Group II (P4, P7); marks a hypo-pigmented phenotypic unit

H = Shannon entropy. Characters with $H < 0.70$ are classified as diagnostic (conserved or rare). Frequency = proportion of the 11 morphs bearing the character.

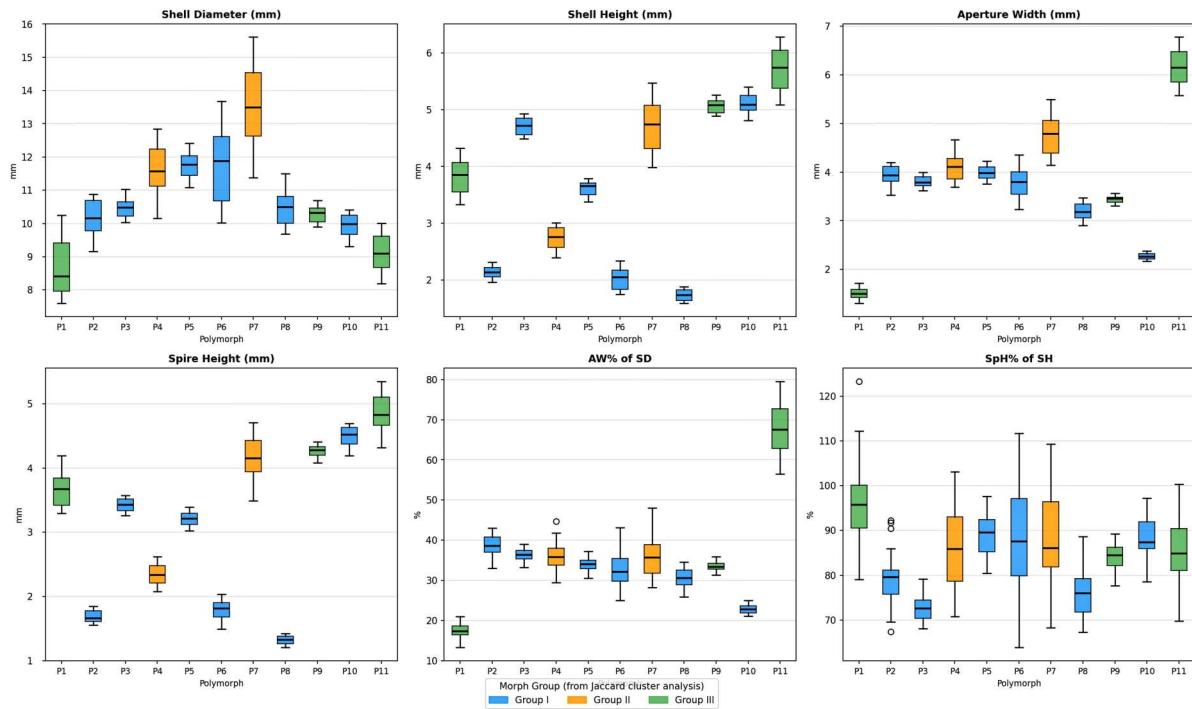


Figure 7. Box-whisker plots of six morphometric parameters across eleven morphs. Boxes coloured by morph group (blue = Group I, orange = Group II, green = Group III). Horizontal line = median; boxes = IQR; whiskers = $1.5 \times \text{IQR}$; circles = outliers.

planispiral form. Group II morphs (P4, P7) combine the largest absolute shell diameters (mean SD = 12.54 mm) with moderate-to-tall shell heights (mean EI = 0.295) and the highest mean aperture width (4.47 mm). Group III morphs (P1, P9, P11) exhibit the smallest shell diameters (mean SD = 9.36 mm) but by far the greatest shell heights (mean EI = 0.524) and widest apertures (mean AW = 3.71 mm), producing a markedly taller, more turbanate shell plan that is architecturally distinct from both other groups (Figure 7).

It should be noted that the morphometric character set comprises four independent linear measurements (shell diameter, shell height, aperture width, spire height); the two additional parameters (AW%SD, SpH%SH) are derived ratios of these same measurements rather than independent characters. This comparatively narrow set of external linear dimensions, while standard for rapid field-based morphometric screening (Marcus 1990; Bookstein 1991), cannot capture finer aspects of shell architecture (whorl profile, aperture shape, shell thickness, ornamentation) that geometric morphometric approaches could resolve, and this should be borne in mind when interpreting the morphometric groupings reported below.

Analysis of variance

One-way ANOVA on the four raw linear measurements confirmed highly significant inter-morph differences across all parameters (all $F_{10,319}$, $p < 0.001$; Table 5). Effect sizes (η^2) were remarkably large, ranging from 0.774 for shell diameter to 0.975 for spire height, indicating that morph identity alone explains 97.5% of total variance in spire height, 96.9% in shell height, and 96.0% in aperture width. All results were confirmed non-parametrically by Kruskal–Wallis tests (all $H > 250$, $p < 0.001$), validating these conclusions independently of normality assumptions. Shapiro–Wilk tests of normality for each morph and parameter group results have been represented in Supplementary Table S1.

Principal component analysis

PCA on standardised raw morphometric data extracted two principal components with eigenvalues > 1.0 , which together explained 83.0% of total morphometric variance (PC1 = 51.1%, PC2 = 31.9%; Figure 8A). Removing derived ratio variables eliminated multicollinearity and yielded orthogonal, clearly interpretable morphological axes. PC1 represents a shell verticality

Table 5. One-way ANOVA and Kruskal-Wallis test statistics for inter-morph morphometric differences ($n = 330$; $df = 10, 319$).

Parameter	F -statistic	df	ANOVA p	KW H	KW p	η^2
Shell Diameter (mm)	108.97	10, 319	< 0.001	256.77	< 0.001	0.774
Shell Height (mm)	1006.66	10, 319	< 0.001	311.16	< 0.001	0.969
Aperture Width (mm)	769.77	10, 319	< 0.001	296.07	< 0.001	0.960
Spire Height (mm)	1263.45	10, 319	< 0.001	315.42	< 0.001	0.975

η^2 = eta-squared (effect size). All p -values < 0.001 by both parametric and non-parametric tests. KW = Kruskal-Wallis H statistic.

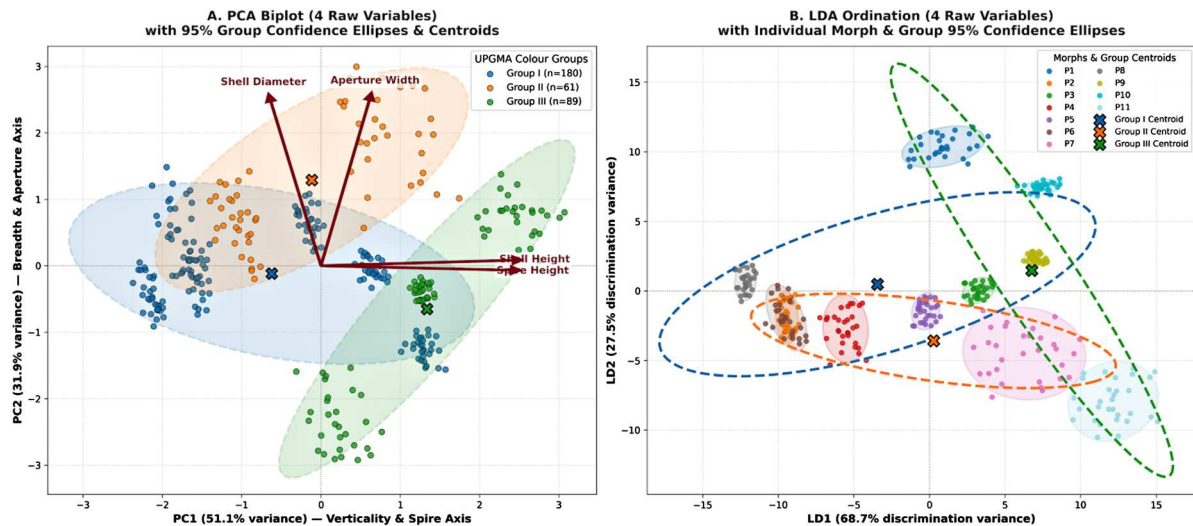


Figure 8. Multivariate Morphometric Ordination of *Umbonium vestiariu* morphs. (A) Principal Component Analysis (PCA) biplot of standardised raw shell measurements ($n = 330$), showing specimen distribution across PC1 (51.1% variance; shell verticality/spire height axis) and PC2 (31.8% variance; shell breadth/aperture axis). Vector arrows indicate variable loadings (Shell Diameter, Shell Height, Aperture Width, Spire Height). Shaded 95% confidence ellipses and centroid markers (X) are plotted for each of the three UPGMA-defined colour-pattern groups (Group I: blue; Group II: orange; Group III: green). (B) Linear Discriminant Analysis (LDA) ordination in LD1–LD2 canonical space (LD1 = 68.7%, LD2 = 27.5% of discrimination variance; combined = 96.2%). Individual specimens are coloured by morph (P1–P11), with clearly delineated 95% confidence ellipses for each individual morph (solid-coloured outlines with tinted shading). Overlaid bold dashed 95% confidence ellipses and centroids (X) represent the broader three UPGMA colour groups, demonstrating fine-scale shape separation among individual morphs alongside partial overlap at the group level.

and spire development axis, dominated by heavy positive loadings for shell height (+0.984) and spire height (+0.976). PC2 captures a shell breadth and aperture size axis, driven by strong positive loadings for aperture width (+0.802) and shell diameter (+0.796). The PCA biplot (Figure 8A), incorporating 95% confidence ellipses around the centroids of the three UPGMA-defined colour groups, visualises their separation in two-dimensional shape space. Group III morphs, especially P11, which attains the highest PC1 positive score; cluster in the upper-right quadrant characterised by tall-spined, vertically extended morphology. Group I morphs cluster in the negative-to-neutral PC1 region, reflecting their depressed, planispiral plan, while Group II morphs project towards high positive PC2 values due to their large absolute shell diameters and wide apertures.

Linear discriminant analysis

LDA applied to the four raw linear measurements achieved effective separation of all eleven morphs in discriminant space (96.4% overall classification accuracy; Figure 8B). LD1 (68.7% of discrimination variance) separates flat, broad-shelled morphs (P2, P6, P8; negative LD1 scores) from tall-shelled forms (P9, P10, P11; positive LD1 scores), driven primarily by spire and shell height loadings. LD2 (27.5% of variance) further discriminates large-aperture morphs (P7, P11) from narrow-aperture forms (P1, P8). The combined LD1 + LD2 solution captures 96.2% of total discrimination variance. Plotting 95% confidence ellipses for each of

the 11 individual morphs demonstrates that individual morphs form discrete, tightly clustered and largely non-overlapping units in canonical space (Figure 8B). However, when bold 95% confidence ellipses are constructed around the centroids of the three UPGMA colour groups, substantial envelope overlap is observed, particularly between Groups I and II. This pattern demonstrates that while raw linear shell dimensions reliably discriminate individual morphs, morphometric variation does not partition cleanly along the broader three-group classification derived from binary colour-pattern characters, reinforcing the discordance between shell coloration and physical architecture.

Group membership (I–III) plotted in the PCA biplot and LDA ordination reflects the a priori colour-pattern classification derived from the Jaccard/UPGMA analysis (UPGMA cluster analysis and morph groups); it is not an independent cluster solution fitted to the morphometric data. The independently derived morphometric clustering (Ward linkage on standardised means) is reported separately and does not reproduce this grouping; a discordance discussed in the synthesis of colour-pattern and morphometric classifications part, the diagnostic characters and their distribution among morph groups part and synthesis: the Sundarban tidal mosaic as a polymorphism engine part.

While individual morphs form tight, largely non-overlapping clusters in both PCA and LDA space (consistent with the discriminant classification success

reported above), 95% confidence ellipses computed around the centroids of the three UPGMA-defined colour groups (Figure 8) show substantial overlap, particularly between Groups I and II. This indicates that although shell shape reliably discriminates individual colour morphs, it does not cleanly partition them into the coarser three-group structure recovered from binary colour-pattern characters, reinforcing the morphometric/colour-pattern discordance documented independently via Ward clustering and discussed in the synthesis of colour-pattern and morphometric classifications part and the synthesis: the Sundarban tidal mosaic as a polymorphism engine part.

Morphometric correlations and growth coupling

Pearson correlation analysis revealed two strong, highly significant positive correlations among morphometric parameters: shell height – spire height ($r = 0.956$, $p < 0.001$) and aperture width – AW%SD ($r = 0.909$, $p < 0.001$; Figure 9A). These tightly coupled relationships reflect integrated growth modules: spire architecture is inseparable from total shell height, and aperture proportions track absolute aperture size, leaving no room for independent variation within these modules. By contrast, shell diameter was weakly correlated with shape parameters ($|r| \leq 0.274$), demonstrating that body-size variation (as expressed in SD) is substantially decoupled from shell shape; lateral and vertical growth axes appear to operate under largely independent regulatory control. The SpH%SH index was largely uncorrelated with other parameters ($|r| \leq 0.287$), indicating it captures a distinct aspect of whorl proportionality not collinear with absolute dimensions.

Hierarchical clustering of mean morphometrics

Ward linkage clustering of standardised mean morphometric profiles (Figure 9B) resolved three groups

at the appropriate cut height: a singleton cluster comprising P11 alone, which splits off at the highest linkage distance (≈ 6.2) and is morphometrically the most divergent morph in the dataset; a cluster comprising P1, P3, P9 and P10; and a cluster comprising P2, P4, P5, P6, P7 and P8. This topology does not correspond to the three-group UPGMA solution derived independently from binary colour-pattern characters: only P4 and P7 (colour Group II) remain grouped together in both classifications, and even this pair is merged into a six-morph cluster with four colour-Group I morphs (P2, P5, P6, P8) in the morphometric analysis. Colour Group I is split across both morphometric clusters (P3, P10 vs. P2, P5, P6, P8), and colour Group III is split three ways (P1, P9 vs. P11 alone). This is visually evident in Figure 9B, where the colour-group tick-label shading does not track the dendrogram branching. We treat this as a substantive finding rather than a technical discrepancy: shell colour pattern and shell morphometry appear to be substantially decoupled in this species (diagnostic characters and their distribution among morph groups part and synthesis: the Sundarban tidal mosaic as a polymorphism engine part).

Distributional analysis across Sundarban Islands

Overall abundance and morph composition

A total of 4033 individuals of *U. vestiarium* were enumerated across 60 quadrats (10 replicates $\times$ 6 islands), spanning six islands of the Sundarban Biosphere Reserve. Island-level totals ranged from 534 individuals at Jamboo Island to 835 at Dhanchi Island, indicating marked spatial heterogeneity in total population density (Figure 10A). The most abundant morph overall was P7 (Group II; $n = 752$; 18.6%), followed by P8 (Group I; $n = 720$; 17.9%) and P1 (Group III; $n =$

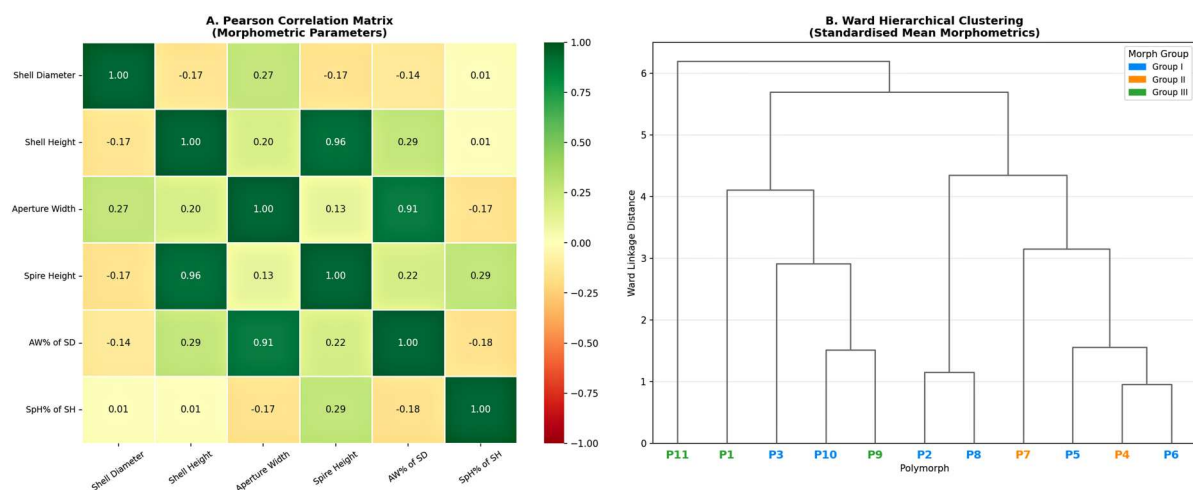


Figure 9. A: Pearson correlation matrix heatmap (green = positive, red = negative). B: Ward linkage dendrogram of 11 morphs based on standardised mean morphometrics. Tick labels coloured by morph group (blue = Group I, orange = Group II, green = Group III).

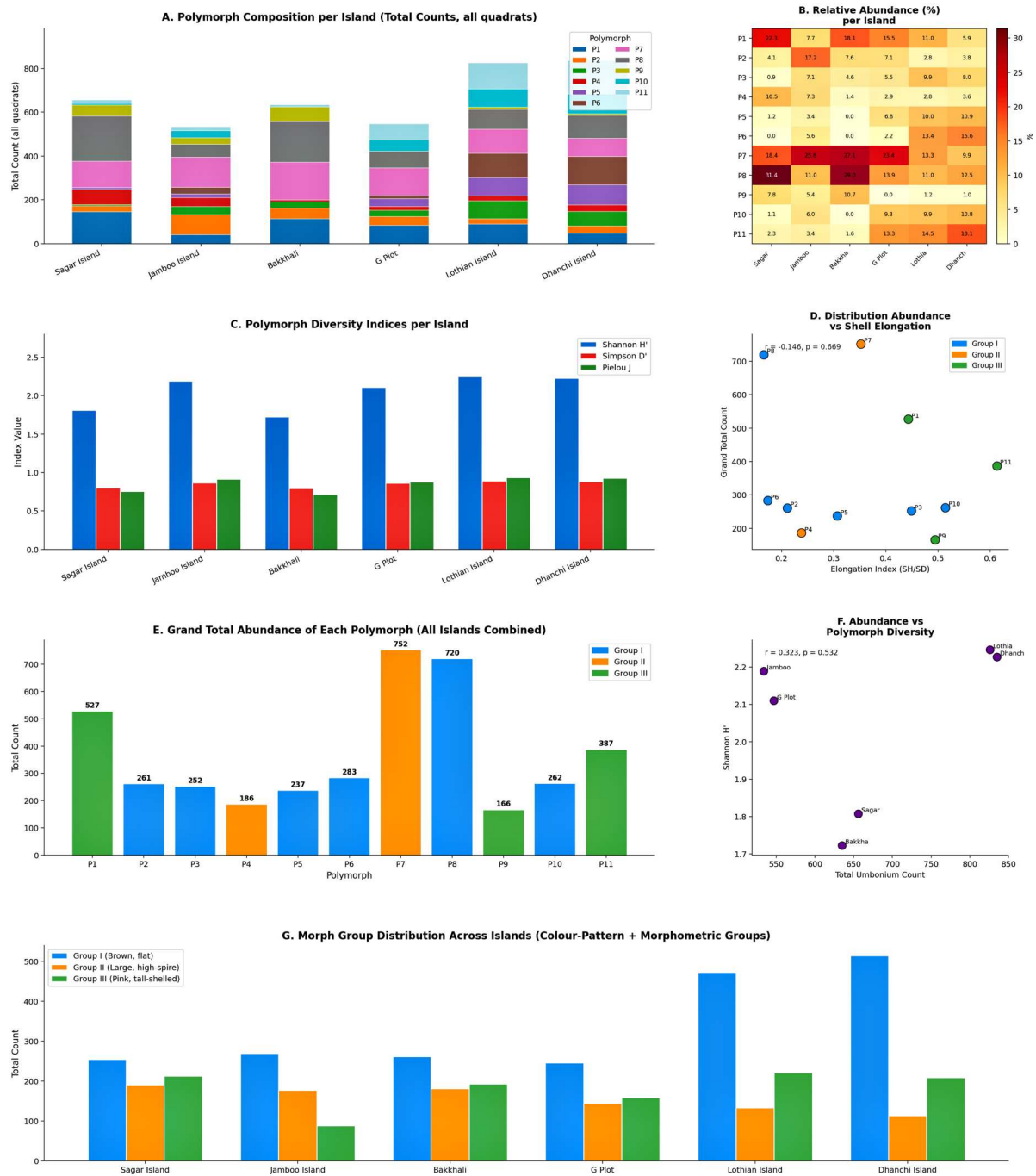


Figure 10. Morph distribution in different islands of Sundarban. (A) Morph composition per island across all quadrats ($n = 10$ replicates per morph per island) (B) Relative abundance (%) of each morph (P1–P11, rows) across all six islands (columns), with warmer colours indicating higher proportional representation. (C) Morph diversity indices per island, including Shannon H' (blue), Simpson D' (red) and Pielou's evenness J (green). (D) Scatter plot of grand total count (distribution abundance) against shell elongation index (SH/SD) for each morph; Pearson correlation coefficient and p -value are annotated ($r = -0.146$, $p = 0.669$). (E) Total abundance of each morph across all islands combined, (F) Total *Umbonium* count per island against Shannon H' diversity index; Pearson r and p -value annotated ($r = 0.323$, $p = 0.532$). (G) Distribution of the three morphological groups (Group I: blue; Group II: orange; Group III: green) in terms of total count across all six study islands.

527; 13.1%). The two rarest morphs were P9 (Group III; $n = 166$; 4.1%) and P4 (Group II; $n = 186$; 4.6%) both notably also the most morphologically distinctive or isolated morphs in the companion clustering analyses (Figure 10E). Individual count from all the quadrates were, Sagar Island: P1 – 146, P2 – 27, P3 – 6, P4 – 69, P5 – 8, P6 – 0, P7 – 121, P8 – 206, P9 – 51, P10 – 7, P11 – 15; Jamboo Island: P1 – 41, P2 – 92, P3 – 38,

P4 – 39, P5 – 18, P6 – 30, P7 – 138, P8 – 59, P9 – 29, P10 – 32, P11 – 18; Bakkhali: P1 – 115, P2 – 48, P3 – 29, P4 – 9, P5 – 0, P6 – 0, P7 – 172, P8 – 184, P9 – 68, P10 – 0, P11 – 10; G plot: P1 – 85, P2 – 39, P3 – 30, P4 – 16, P5 – 37, P6 – 12, P7 – 128, P8 – 76, P9 – 0, P10 – 51, P11 – 73; Lothian Island: P1 – 91, P2 – 23, P3 – 82, P4 – 23, P5 – 83, P6 – 111, P7 – 110, P8 – 91, P9 – 10, P10 – 82, P11 – 120; Dhanchi Island: P1 – 49,

P2 – 32, P3 – 67, P4 – 30, P5 – 91, P6 – 130, P7 – 83, P8 – 104, P9 – 8, P10 – 90, P11 – 151.

Non-Random distribution: chi-square analysis

The chi-square contingency test applied to the 11×6 morph $\times$ island count matrix yielded $\chi^2 = 1463.99$ (df = 50, $p < 0.0001$), decisively rejecting the null hypothesis of random morph distribution across islands. The effect size statistic Cramér's $V = 0.425$ indicates a moderate-to-strong association between morph identity and island of occurrence; stronger than would be anticipated even allowing for the extremely short pelagic phase of the larvae of *U. Vestiarium*. Examination of standardised residuals from the chi-square test identifies the sharpest non-random patterns: P7 was markedly over-represented at Bakkhali (observed $n = 172$ vs. expected ~ 116) and under-represented at Dhanchi (83 vs. expected ~ 131); P6 was strongly concentrated at Dhanchi and Lothian Islands (combined $n = 241$) while recording zero individuals at both Sagar Island and Bakkhali; and P9 was completely absent from G Plot across all 10 replicates. These patterns collectively indicate that island-specific microhabitat selection acts differentially on colour-pattern morphotypes, providing direct field evidence for spatially variable (mosaic) selection (Figure 10B).

Inter-Island abundance variation: ANOVA and Kruskal-Wallis

One-way ANOVA revealed highly significant inter-island abundance differences for all 11 morphs (all $F_{5,54}$, $p < 0.001$), with F -statistics ranging from 55.28 (P7) to 361.79 (P6; Figure 10). Morph 6 registered the largest F -statistic of the entire dataset ($F = 361.79$), driven by its extreme concentration at Lothian and Dhanchi Islands and complete absence from Sagar Island and Bakkhali, the sharpest habitat specificity observed for any morph in this study. In stark contrast, morph 7 produced the lowest F -statistic (55.28), reflecting its comparatively even distribution across all six islands and identifying it as the dominant cosmopolitan morph of the Sundarban population. All ANOVA results were independently confirmed by Kruskal–Wallis ($H = 46.03$ – 55.58 ; all $p < 0.001$), validating the significance of inter-island variation without parametric assumptions (Figure 10A, B, G).

Morph diversity indices per island

Shannon diversity (H') and associated indices are presented in Table 6. Morph diversity was highest at the outer, more marine-influenced islands: Lothian Island ($H' = 2.247$; Pielou's $J = 0.937$) and Dhanchi Island ($H' = 2.227$; $J = 0.929$) both harboured the full complement of all 11 morphs. Jamboo Island also supported all 11 morphs with high evenness ($H' = 2.190$; $J = 0.913$). At the other extreme, Bakkhali recorded the lowest diversity ($H' = 1.723$; $S = 8$; $J = 0.719$), with P5, P6 and P10 entirely absent and the community heavily dominated by P7 and P8. Simpson's index (D') ranged from 0.791 (Bakkhali) to 0.888 (Lothian Island), confirming that the probability of encountering two individuals of the same morphotype is lowest at the outer islands. A positive rank correlation between total island abundance and Shannon H' (Spearman $\rho = 0.714$) suggests that higher-density populations support greater morph diversity, consistent with density-dependent apostatic selection maintaining rare morphs (Figure 10C, F).

Integration of distribution, morphology and morphometry

Table 7 synthesises the distributional, morphological and morphometric profiles of all 11 morphs, enabling cross-framework ecological guild assignment. Three coherent ecological patterns emerge from this integration, each corresponding to a morph group.

Group I morphs (P2, P3, P5, P6, P8, P10) collectively account for 49.9% of total abundance. Their low elongation indices ($El = 0.17$ – 0.31) and broad shell diameters (mean $SD = 10.74$ mm) are consistent with a depressed, planispiral shell form adapted to high-intertidal, exposed sandy flats. Within this group, P8 and P2 peak at Sagar Island and Bakkhali (exposed, moderate-to-high wave energy), while P6 achieves its near-exclusive distribution at Lothian and Dhanchi, illustrating fine-scale sub-group micro-habitat partitioning within a single colour-pattern cluster.

Group II morphs (P4, P7) together represent 23.3% of abundance. P7, the overall dominant ($n = 752$; $El = 0.352$; lowest ANOVA $F = 55.28$), has the largest shell diameter of all morphs (mean $SD = 13.52$ mm) and occurs across all six islands, qualifying it as a habitat generalist of the Sundarban intertidal zone. P4, by contrast, is the most morphologically isolated morph in

Table 6. Morph diversity indices per island. H' = Shannon diversity; D' = Simpson's index; J = Pielou's evenness. S = morph richness.

Island	Total N	Richness (S)	Shannon H'	Simpson D'	Pielou J	Interpretation
Sagar Island	656	10	1.808	0.799	0.754	Moderate diversity; P8 dominant; P6, P9 rare
Jamboo Island	534	11	2.190	0.865	0.913	High; all 11 morphs; most even distribution
Bakkhali	635	8	1.723	0.791	0.719	Lowest; P5, P6, P10 absent; P7, P8 peak
G Plot	547	10	2.110	0.863	0.880	High; P9 absent; centrally positioned diversity
Lothian Island	826	11	2.247	0.888	0.937	Highest; all morphs; P6, P11 codominant
Dhanchi Island	835	11	2.227	0.883	0.929	High; P6, P11 dominant; most abundant island

All diversity values computed using morph counts as the diversity units across 10 replicate quadrats per island. Lothian and Dhanchi islands record both the highest total abundances and the highest diversity values.

Table 7. Integrated summary of morph-level distribution, key morphometric parameters, Jaccard mean similarity and ecological guild assignment across 11 *Umbonium vestiarium* morphs.

P	Grp	Total	Rank	Primary Islands	SD (mm)	EI	AW%SD	Ecological Guild
P1	III	527	1	Sagar, Bakkhali	86.64	0.443	17.42	Moderate intertidal, sheltered
P2	I	261	7	Jamboo Island	101.64	0.211	38.72	Moderate-energy sandy flat
P3	I	252	8	Lothian, Dhanchi	104.88	0.449	36.28	Southern island specialist
P4	II	186	10	Sagar Island	115.59	0.238	35.86	Micro-habitat specialist (most isolated)
P5	I	237	9	Lothian, Dhanchi	117.77	0.307	33.90	Low-energy southern specialist
P6	I	283	5	Lothian, Dhanchi	117.55	0.174	32.62	Extreme southern specialist
P7	II	752	1	All islands	135.27	0.352	35.58	Cosmopolitan dominant morph
P8	I	720	3	Sagar, Bakkhali	104.88	0.166	30.44	High-intertidal flat specialist
P9	III	166	11	Sagar, Bakkhali	102.97	0.494	33.29	Sandy flat, restricted range
P10	I	262	6	Lothian, Dhanchi	99.50	0.514	22.75	Low-energy southern specialist
P11	III	387	4	Dhanchi, Lothian	91.10	0.613	68.10	Sheltered intertidal; tall-shelled

SD = shell diameter; EI = elongation index; AW%SD = aperture width as percentage of shell diameter; Jaccard avg = mean pairwise Jaccard similarity of each morph versus all others. Ecological guild assignments are based on convergent evidence from all three analytical frameworks.

the dataset (mean Jaccard $J=0.216$) and is largely restricted to Sagar Island. Its narrow distributional range and unique character suite together suggest a micro-endemic or habitat-specialist status warranting conservation attention.

Group III morphs (P1, P9, P11) combine moderate-to-low total abundance (26.8%) with the highest elongation indices ($EI=0.44$ – 0.61) and widest apertures (AW up to 6.18 mm in P11), suggesting a taller, more turbinated shell architecture associated with sheltered or lower-energy intertidal micro-environments. P11, simultaneously the most morphometrically extreme ($EI=0.613$) and chromatically unique (grey-on-grey surface; autapomorphic Shannon entropy = 0.44) morph, reaches its highest densities at Dhanchi and Lothian islands, supporting the inference of a distinct low-energy habitat preference for this morphotype.

Synthesis: discordance between colour-pattern and morphometric classifications

Taken together, the results of the four analytical frameworks; taxonomic description, morphological cluster analysis, shell morphometry and island-based distributional analysis, present a convergent, multi-dimensional portrait of intraspecific polymorphism in *U. vestiarium* within the Sundarban Biosphere Reserve. Taken together, the four analytical frameworks used here; taxonomic description, colour-pattern cluster analysis, shell morphometry and island-based distributional analysis, do not converge on a single consistent partition of the eleven morphs. Colour-pattern clustering and island-level distributional guild patterns are broadly consistent with one another (distributional analysis across Sundarban Islands part), but morphometric clustering identifies a substantially different grouping (hierarchical clustering of mean morphometrics part). This decoupling of colour pattern from shell shape is itself an informative result: it suggests that colour and morphometric variation may be governed by

partly independent developmental, genetic, or environmental processes, rather than co-varying as a single integrated phenotype. The identification of rare, morphologically isolated morphs (P4, P9) with restricted island distributions continues to highlight the conservation significance of phenotypic diversity monitoring in this globally important coastal ecosystem, independent of the mechanism ultimately shown to underlie it.

Discussion

Adaptive divergence: evidence and limitations

Before interpreting the patterns reported here, it is necessary to state explicitly what would be required to demonstrate that the observed polymorphism reflects adaptive divergence, and to assess how far the present dataset meets those requirements. Three conditions must be satisfied. First, at least part of the phenotypic variation described here, colour pattern, shell morphometrics, or both must be shown to have a genetic basis, rather than being wholly attributable to environmentally induced plasticity or pseudo-polymorphism (Grüneberg 1980). Second, the non-random distribution of phenotypes among islands must be shown to reflect a non-random distribution of the underlying genetic variants, and this spatial genetic structure must be distinguished from structure generated by the stochastic extinction and recolonisation of local populations rather than by selection. This distinction is particularly important in the Sundarban Biosphere Reserve, where islands are subject to ongoing erosion, accretion and shifting mangrove-substrate composition (Sahana *et al.* 2016, 2022; Samanta *et al.* 2021), and where a short (~48-hour) planktonic larval duration (Berry 1987) limits dispersal and could permit island populations to diverge through founder effects and genetic drift independent of any selective process. Third, the selective factors proposed here, thermoregulatory selection, substrate matching and density-dependent apostatic selection,

must be shown to directly influence the frequency of the genetic variants underlying the phenotype, rather than merely correlating with its spatial distribution.

The present study does not satisfy any of these three conditions. No molecular genetic data were collected: the binary character and Jaccard dissimilarity framework used to define morph groups is a phenotypic classification tool and cannot establish whether morph groups correspond to discrete genetic polymorphisms, environmentally induced pseudo-polymorphism, or a mixture of both (Grüneberg 1980). This concern applies with particular force to the morphometric dimensions measured here. Spire height and aperture width; the two parameters most responsible for separating morph groups in the present analysis are among the shell traits most consistently documented as environmentally plastic in gastropods: shell allometry tracking individual growth rate rather than genotype (Kemp & Bertness 1984), and shorter-spined, wider-apertured shells developing as a direct response to wave exposure, independent of any genetic difference (Trussell *et al.* 1993). The wave-exposure gradient invoked in the shell morphometric differentiation among colour-defined morphs part to interpret morph-group differences in elongation index and aperture width is therefore equally consistent with a purely plastic response of individual snails to local hydrodynamic conditions as with genetically based adaptive divergence, and the present phenotype-only dataset cannot distinguish between these alternatives. Consequently, the island-level distributional differences reported in distributional analysis across Sundarban Islands part cannot presently be attributed to selection acting on genetic variants as opposed to differential colonisation history, genetic drift among islands with limited larval exchange, or stochastic local extinction and recolonisation; a distinction that would require population genetic markers (e.g., microsatellites or RADseq) to test for isolation-by-distance, effective migration rates and island-level genetic differentiation. Nor were any of the proposed selective mechanisms tested directly: no reciprocal transplant experiments, predation assays, or thermal performance trials were conducted, and no data on survival or reproductive fitness of different morphs under different habitat conditions were collected. The present study is therefore a single-timepoint, phenotype-only, correlational dataset. It can establish that the eleven morphs form three reproducible, non-random phenotypic and distributional groupings (synthesis of colour-pattern and morphometric classifications part); it cannot, on its own, establish that these groupings arose through, or are maintained by, natural selection. The remainder of this Discussion interprets the observed patterns against this constraint, and proposes the mosaic adaptive divergence

model set out in synthesis: the Sundarban tidal mosaic as a polymorphism engine part as a hypothesis motivated by, rather than confirmed by, the data, to be tested through the molecular and experimental approaches outlined below.

Two further limitations bear on the temporal generality of the findings. The present study represents a single temporal snapshot of morph frequencies; Sivadas *et al.* (2012) documented significant seasonal variation in *U. vestiarium* abundance and size structure in Indian populations, and whether Sundarban morph frequencies fluctuate seasonally through differential recruitment, selective mortality, or ontogenetic colour change (Grüneberg 1980) cannot be resolved from the present data. Longitudinal monitoring across multiple time points within the annual cycle, and ideally across multiple years, is necessary to assess the temporal stability of the morph-group and island-distributional patterns reported here, and would itself provide an indirect test of the extinction–recolonisation alternative raised above: stable inter-annual morph composition on a given island would argue against a purely stochastic colonisation history, whereas high inter-annual turnover would support it.

Addressing these gaps will require an integrated program combining population genetic analysis (to test for genetic differentiation and its correspondence to phenotype), pigment biochemical analysis via spectrophotometry and HPLC (to determine whether morph-group pigmentation differences reflect distinct, plausibly genetically-controlled biochemical pathways such as melanin, carotenoid, or polyene chromophores, versus continuously variable pigment expression), and field experiments like reciprocal transplants and predation assays, to test whether the proposed selective agents measurably affect survival or reproduction of different morphs *in situ*.

An additional methodological limitation of the present dataset concerns the subsampling procedure used to select the 30 representative specimens per colour morph for linear morphometric analysis. Although specimens were drawn blindly after excluding damaged shells (>20% surface damage) to avoid conscious investigator bias towards larger or more striking individuals, a formal mathematical randomisation protocol (e.g., cataloguing all 4033 individuals and applying a random-number generator) was not implemented. Consequently, we cannot strictly rule out the possibility that subtle convenience or handling biases influenced the size distribution of the measured subset. However, given that one-way ANOVA and Kruskal–Wallis tests revealed extraordinarily high effect sizes across all four linear dimensions ($\eta^2 = 0.774\text{--}0.975$) and Linear Discriminant Analysis achieved 96.4% classification accuracy across morphs, it is highly improbable that the distinct three-

dimensional architectural profiles observed among morph groups are artefacts of subsampling bias. Nonetheless, future morphometric investigations should implement formal randomised block sampling across island replicates to ensure that within-morph size allometry is fully and impartially represented.

Morph diversity in geographic context

The eleven colour morphs of *U. vestiarium* documented from the Sundarban Biosphere Reserve place this population comfortably within the broader gradient of chromatic diversity now recorded across the species' Indo-Pacific range. Comparing this figure against published accounts, four morphs from Bangladesh (Hossain & Maxwell 2022), eleven from Kalbadevi Bay, western India (Sivadas *et al.* 2012), up to twenty-four from Parangipettai on India's southeastern coast (Sivadas *et al.* 2012), and a remarkable ninety-six from Singapore (Chan *et al.* 2016); it is clear that morph richness in this species is strongly locality-specific rather than uniformly expressed. The Sundarban count of eleven is with the same as the western Indian figure. That the geographically nearest studied population in Bangladesh yields only four morphs (Hossain & Maxwell 2022) is intriguing, and likely reflects either sampling differences or genuine habitat-driven reductions in morph richness rather than a biological discontinuity.

These findings must be interpreted against the foundational framework of Grüneberg (1980), whose analysis of approximately 38,000 individuals across eleven Indo-Pacific localities remains the most comprehensive phenotypic treatment of *Umbonium* variation to date. Grüneberg (1980) identified two coexisting categories of shell variation: discrete, sharply classifiable patterns likely governed by major-effect genes (true polymorphism), and a majority of forms merging continuously into one another; a condition he termed 'pseudo-polymorphism', wherein a single genotype may produce several phenotypes depending on environmental and developmental conditions. The Sundarban morphs exhibit both characteristics. Two candidate character systems; band position (essentially binary: top-of-whorl in ten of eleven morphs versus bottom-of-whorl, exclusive to P8) and band colour ('Band: Brown', present in 82% of individuals); show the lowest Shannon entropy in the dataset and are the strongest candidates, on frequency grounds alone, for a comparatively simple genetic basis, in contrast to the sixteen higher-entropy characters ($H \geq 0.70$) that more plausibly reflect the continuous phenotypic merging Grüneberg (1980) attributed to pseudo-polymorphic developmental plasticity. However, low entropy in a phenotypic frequency distribution cannot, on its own, distinguish a genetically canalised character from an environmentally canalised

one: in the congeneric trochid *Austrocochlea constricta*, shell-banding pattern, the same class of character treated here as a candidate genetic marker has been shown experimentally to be strongly influenced by diet and habitat, with transplanted individuals shifting banding pattern towards that typical of their new location (Creese & Underwood 1976). We therefore treat band position and band colour as candidate, rather than confirmed, genetically influenced characters; distinguishing a genetic from an environmental basis for these characters would require controlled rearing or transplant experiments analogous to those used for *Austrocochlea*, or direct genetic marker analysis, neither of which was undertaken here.

Morphological divergence and the three-group architecture

The UPGMA cluster analysis, based on pairwise Jaccard dissimilarities across 25 binary characters, resolved a coherent three-group architecture. The overall mean pairwise Jaccard similarity of 0.272 ± 0.113 indicating that any two randomly selected morphs share fewer than a third of their character states reflects a remarkably high degree of intraspecific phenotypic differentiation. Although the Sundarban sample spans only eleven morphs compared to the ninety-six documented by Chan *et al.* (2016) from Singapore, the breadth of inter-morph character divergence ranging from near-complementary pairs (Jaccard similarity: 0.059, P8–P11) to highly similar pairs (0.800, P2–P8) demonstrates that phenotypic diversity in this population is qualitatively rich rather than merely quantitatively restricted.

Group I (P2, P3, P5, P6, P8, P10), the most abundant assemblage accounting for 49.9% of all enumerated individuals, is united by brown banding and near-universal conservation of the band-at-top-of-whorl character. The internal sub-structure revealed at the four-group level sub-cluster Ia (P2, P3, P8) favouring inner islands and sub-cluster Ib (P5, P6, P10) concentrated on the outer southern islands of Lothian and Dhanchi is ecologically interpretable in the context of the Sundarban's pronounced environmental gradients in substrate composition, wave exposure and salinity (Sahana *et al.* 2022; Samanta *et al.* 2021). This fine-scale within-group spatial differentiation is consistent with landscape-level spatially variable selection operating across the tidal mosaic.

Group II (P4, P7), the smallest and most morphologically cohesive unit, is uniquely characterised by uniform stripes combined with translucent pattern expression; a combination absent from all other morphs in this population. The translucent phenotype, involving substantially reduced pigment deposition and a semi-glassy shell appearance, represents the most extreme form of light colouration in the

Sundarban population. This is directly relevant to thermoregulatory hypotheses for gastropod colour polymorphism: Chan *et al.* (2016) proposed that lighter-coloured morphs carry a selective advantage in high-insolation tropical intertidal habitats by avoiding overheating, a mechanism well-supported for littorinid snails. Despite sharing a broadly similar chromatic phenotype, the contrasting abundances of P7 (the most cosmopolitan morph; $n = 752$) and P4 (a restricted specialist; $n = 186$) demonstrate that a shared colour pattern does not guarantee ecological equivalence, and that secondary morphometric and distributional characters are essential for fully characterising each morph's ecological role (Gefaell *et al.* 2023).

Group III (P1, P9, P11), united by pink banding and wavy-line ornamentation, segregates cleanly from the brown-banded majority in both cluster analysis and PCoA ordination. Pink and red shell colouration in marine gastropods is typically attributable to carotenoid pigments, haem-based compounds, or polyene chromophores rather than melanin-based pathways responsible for brown pigmentation (Grüneberg 1980). Grüneberg's (1980) detailed description of pink forms across Indian and Ceylonese populations noted that pink pigment grades continuously from brilliant colour to near-invisibility; a pattern inconsistent with a single discrete allele and more consistent with polygenic or epigenetic regulation. The Group III morphs of the Sundarban population appear to represent a phenotypically stabilised subset of this broader continuum, and their ecological association with sheltered, lower-energy intertidal environments (Lothian and Dhanchi for P11; Sagar and Bakkhali for P1 and P9) suggests a probability of habitat-specific pigmentation selection consistent with background colour matching in intertidal molluscs (Rodrigues & Absalão 2005; Miura *et al.* 2007).

Diagnostic characters and their distribution among morph groups

The frequency and entropy patterns described below are used purely descriptively, to characterise how widely each character state is shared among morphs; they do not, on their own, establish which characters are under genetic versus environmental control, and the cladistic-sounding contrasts drawn here (e.g., between widely shared and narrowly restricted states) should be read as descriptions of phenotypic distribution, not as claims about heritability or evolutionary derivation.

The Shannon entropy analysis provides a quantitative basis for distinguishing characters of high taxonomic value from those reflecting background plasticity. 'Band position: top of whorl' is the most conserved character ($H = 0.44$, frequency = 0.91) and is both ecologically and developmentally interpretable:

dorsally positioned pigment bands in the UV- and thermally stressed apical region may confer photoprotective advantages through melanin-based reduction of UV penetration to the underlying mantle tissue. Its near-universal occurrence across ten of eleven morphs qualifies it as the most widely shared diagnostic character for *U. vestiarium* in the Sundarbans, while its absence in P8 alone confers notable restricted distribution on that morph's basal band position.

The grey basal colouration restricted to P11 is likely arising from partial dilution of dark pigmentation, possibly representing an extreme expression of Grüneberg's (1980) 'whitewash' phenotype is particularly compelling when considered alongside P11's concurrent morphometric distinctiveness (highest elongation index: $El = 0.613$; widest aperture: $AW = 6.18$ mm). P11 is therefore unusual in showing joint chromatic and morphometric distinctiveness, consistent with its status as a singleton outlier in the Ward morphometric clustering. However, this single-morph pattern should not be read as representative of the relationship between colour and shape more generally: across the remaining ten morphs, colour-pattern and morphometric classification are substantially decoupled, and P11's dual distinctiveness is better regarded as a notable individual case than as evidence of a general co-divergence pattern.

Finally, the classification of 'Band: Brown' as a most widespread state ($H = 0.68$, frequency = 0.82) is consistent with its near-universal dominance across all published *U. vestiarium* population studies (Grüneberg 1980; Sivadas *et al.* 2012; Chan *et al.* 2016; Hossain & Maxwell 2022); implies that the pink pigmentation of Group III and the translucent hypo-pigmentation of Group II both represent restricted distribution, potentially maintained by different selective mechanisms: habitat-specific background matching in the former and thermoregulatory advantage in the latter. While necessarily provisional without genetic data, this polarity inference provides a tractable hypothesis for future molecular investigations into the pigmentation genetics and micro-evolutionary history of this ecologically significant intertidal species.

Shell morphometric differentiation among colour-defined morphs

Linear discriminant analysis achieved effective separation of all eleven colour-defined morphs in morphometric space, and ANOVA confirmed highly significant differences in shell dimensions among them. This is a distinct result from, and should not be conflated with, the unsupervised Ward clustering result reported in the hierarchical clustering of mean morphometrics part: LDA is a supervised technique that tests whether individuals already assigned to colour-based morph categories differ in shape, and

by construction can achieve strong separation even when the natural, unsupervised structure of the morphometric data does not reproduce the same group boundaries. Here, that is exactly what occurs; the eleven morphs are morphometrically distinguishable as individual units (LDA), but when morphometric data alone are clustered from scratch (Ward), the resulting three-group partition does not match the colour-based three-group partition. We therefore conclude that shell morphometric variation among morphs is real, substantial, and not attributable to measurement noise, but that it is not organised along the same group boundaries as colour pattern. This decoupling is inconsistent with a model in which colour and shape diverge as a single, co-varying phenotypic module, and is discussed further in the synthesis part of the result and discussion.

The ecological interpretations below use colour-defined group membership (Groups I–III) as the unit of comparison, consistent with how morphs were originally classified, rather than the independent Ward morphometric clusters, which do not align with the colour groups. Within this colour-based framework, the morphometric differences among groups are ecologically interpretable and align coherently with the distributional patterns observed. Group I morphs (mean elongation index, $EI = 0.279$) display the broad, depressed, planispiral shell form typical of high-intertidal sandy flat dwellers: low-profile shells reduce mechanical drag on mobile substrates, minimise desiccation surface area, and facilitate rapid locomotion through the uppermost sediment layer (Kalayanasundaram *et al.* 1974; Tamaki & Kikuchi 1983). This interpretation is reinforced by Group I's distributional dominance at Bakkhali and Sagar Island; broad, exposed beaches with moderate to high wave energy. In contrast, Group III morphs (mean $EI = 0.524$), with taller turritate shells and exceptionally wide apertures (P11: aperture width = 6.18 mm; $AW\%SD = 68.1\%$), are biomechanically consistent with deeper intertidal or semi-sheltered habitats where desiccation pressure is reduced, concordant with Group III's preference for the outer, more stable Lothian and Dhanchi islands. We note that this correspondence between shell shape and wave-exposure regime is consistent with an adaptive interpretation, but is equally consistent with direct developmental plasticity in response to local hydrodynamic conditions, a well-documented phenomenon in other gastropods (Kemp & Bertness 1984; Trussell *et al.* 1993). Distinguishing these possibilities would require the common-garden or reciprocal-transplant experiments outlined in adaptive divergence: evidence & limitations part. Group II morphs (mean $EI = 0.295$) occupy an intermediate morphometric position but achieve the largest absolute shell diameters (mean $SD = 12.54$ mm), particularly P7, whose broadly generalist body plan may

confer adequate performance across diverse intertidal conditions.

The strong covariation (Pearson 1896) between shell height and spire height ($r = 0.956$, $p < 0.001$) and between aperture width and $AW\%SD$ ($r = 0.909$, $p < 0.001$) reveals integrated growth modules within the shell architecture, consistent with conserved spire-development programs reported for other trochid gastropods (Gould 1966; Rohlf & Marcus 1993). Conversely, the near-independence of shell diameter from shape parameters ($|r| \leq 0.274$) indicates that body size and shape are governed by partially decoupled developmental mechanisms, allowing them to evolve semi-independently across morphs. Critically, the extraordinarily high eta-squared values for shell height ($\eta^2 = 0.969$), aperture width (0.960), and spire height (0.975) indicate that morph identity alone accounts for 96–97% of total variance in these parameters, confirming that morphometric differences among morphs are large, highly consistent and not attributable to measurement error or sampling noise. Whether this consistency reflects genetic control, environmental canalisation, or a combination of both cannot be determined from the present phenotype-only dataset, and we do not interpret it as evidence of genetic determination without the molecular or experimental data that would be needed to establish this.

Non-random morph distribution and the mosaic selection hypothesis

The chi-square analysis ($\chi^2 = 1463.99$, $df = 50$, $p < 0.0001$; Cramér's $V = 0.425$) unequivocally establishes that morph distribution across the six Sundarban islands is highly non-random. A Cramér's V approaching 0.5 is the conventional threshold for a large effect and is remarkable for a geographically compact archipelago and strongly indicates that island identity is a meaningful predictor of morph composition. In the framework of polymorphism theory (Ford 1965; Leimar 2005, 2009), non-random spatial distribution provides necessary, if not alone sufficient, evidence for spatially variable selection. Crucially, all eleven morphs individually show significant inter-island abundance variation by both ANOVA ($F = 55.28$ – 361.79) and Kruskal–Wallis ($H = 46.03$ – 55.58 ; $p < 0.001$ in every case), transforming the overall pattern into a morph-specific signal meaning each morph carries a characteristic island-level distribution profile.

Crucially, all eleven morphs individually show significant inter-island abundance variation by one-way ANOVA ($p < 0.001$ in every case), transforming the overall pattern into a morph-specific signal wherein each morph carries a characteristic island-level distribution profile. Because our sampling effort was identically balanced across all morphs ($df = 5, 54$), the

magnitude of the ANOVA F -ratio directly quantifies the proportion of spatial variance explained by island identity, providing a robust biometric index of spatial specialisation (Sokal & Rohlf 1995; Underwood 1997). The most extreme case is morph 6, with an F -statistic of 361.79, which is the highest in the dataset, driven by complete absence from Sagar Island and Bakkhali and overwhelming dominance at Lothian and Dhanchi. Combined with P6's lowest elongation index of any morph ($El=0.174$) and its flat, large-shelled morphometry, this points strongly to extreme substrate specialisation in fine-grained, low-energy, high-salinity outer island environments; conditions known to favour depressed trochid gastropods in Indo-Pacific intertidal settings (Tamaki & Kikuchi 1983; Noda *et al.* 1995). At the opposite extreme, morph 7's comparatively modest F -statistic (55.28) and presence across all six islands reflect the generalist habitat tolerance associated with its large body size and intermediate elongation index, making it the closest ecological equivalent to a habitat generalist within the Sundarban *Umbonium* community.

The complete absence of specific morphs from entire islands; P6 from Sagar and Bakkhali, P5 and P10 from Bakkhali, and P9 from G Plot is unlikely to represent a sampling artefact given consistent zero counts across all ten replicates. These patterns indicate genuine local exclusion of certain morphotypes from islands whose environmental conditions fall outside their ecological tolerance range. In the context of Gröneberg's (1980) distinction between true and pseudo-polymorphism, geographically restricted distributions of this kind argue in favour of a genetic basis for the character combinations defining excluded morphs, since purely plastic responses to local conditions would predict more cosmopolitan distributions. These findings are also consistent with Chan *et al.* (2016) observation that, because the principal predators of *U. vestiarium* are naticid and olivid gastropods and sea stars, which hunt primarily by chemotaxis rather than visual search, classical apostatic selection based on visual search-image formation is unlikely to be the dominant maintenance mechanism; instead, spatially heterogeneous selection mediated by substrate matching, thermal regime and hydrodynamic conditions most plausibly generates the island-specific morph signatures observed here.

Morph diversity, island ecology and density-dependent processes

Morph diversity across the six islands ranges from $H' = 1.723$ at Bakkhali ($S=8$ morphs) to $H' = 2.247$ at Lothian Island ($S=11$ morphs), aligning with a broader pattern wherein outer, marine-influenced islands support greater morph richness and evenness than inner, riverine-influenced islands. This gradient

has two non-exclusive explanations. First, the more stable, high-salinity outer islands likely offer greater microhabitat heterogeneity having finer and more variable sediment gradients, more diverse reflectance substrates and richer predator communities supporting a wider range of ecologically viable phenotypic niches. Second, the positive Spearman rank correlation between total island abundance and Shannon H' ($\rho = 0.714$) suggests that higher-density populations sustain greater morph diversity, consistent with density-dependent apostatic selection wherein the fitness advantage of rare morphs increases as common morphs accumulate (Ekendahl 1998; Rodrigues & Absalão 2005). Under this model, the high-density populations of Lothian ($n=826$) and Dhanchi ($n=835$) would be expected to maintain rare morphs at higher frequencies, these are precisely the observed patterns, and one consistent with density-diversity relationships documented in other polymorphic gastropods (Sivadas *et al.* 2012).

The low diversity and reduced evenness at Bakkhali ($H' = 1.723$; Pielou's $J = 0.719$) reflects the complete absence of P5, P6, and P10; Group I morphs with outer-island preferences from this exposed, moderately high-energy beach. The dominance of P7 and P8 at Bakkhali, both associated with exposed dynamic sandy flats, suggests that this island's physical regime acts as an environmental filter, selectively admitting morphs suited to high-energy conditions. Environmental filtering of intertidal macrobenthic assemblages following physical disturbance of sandy substrates is well-documented (Ong & Krishnan 1995), and the morph exclusions observed at Bakkhali are consistent with this mechanism operating at the level of phenotypic variant rather than species. This environmental filtering hypothesis is mechanistically distinct from but not mutually exclusive with apostatic and substrate-matching selection, and their combined operation within the Sundarban tidal mosaic most plausibly generates the complex, multi-layered distributional signatures observed across the six islands.

Mechanisms maintaining polymorphism: towards a synthetic model

Taken together, the morphological, morphometric, and distributional findings of this study are most parsimoniously explained by a mosaic adaptive divergence model, wherein multiple selective mechanisms operate simultaneously across the spatially heterogeneous Sundarban landscape. No single mechanism suffices to account for the full pattern; rather, the complex environmental mosaic of the SBR, encompassing salinity gradients, substrate heterogeneity, wave energy regimes, and thermal variability (Sahana *et al.* 2016; Samanta *et al.* 2021) generates a spatial patchwork of distinct selective environments, each

favouring different morphs. The three morph groups consistently recovered across all analytical frameworks correspond to three recognisable ecological guilds: a sandy-flat guild (Group I) adapted to exposed high-intertidal environments; a generalist-to-specialist guild (Group II) spanning from cosmopolitan dominance (P7) to microhabitat restriction (P4); and a sheltered intertidal guild (Group III) characterised by tall turbanate shells and pink pigmentation associated with lower-energy outer island conditions.

The thermoregulatory selection hypothesis, proposing that lighter, less pigmented shells experience lower thermal loading in high-insolation tropical intertidal habitats (Miura *et al.* 2007; Chan *et al.* 2016), receives only partial and equivocal support from the island-level distribution of pale and translucent morphs recorded here. Rather than converging on a shared set of high-insolation islands, the palest morphs diverge sharply in distributional breadth: P7 occurs as a habitat generalist across all six islands irrespective of presumed thermal regime, while P4 and P6 are each restricted to a small, non-overlapping subset of islands. This is not the pattern a straightforward thermoregulatory-selection model would predict, and suggests that, if thermal loading contributes to the distribution of pale morphs, it does so unevenly and jointly with other factors (substrate, salinity, wave exposure) rather than as a single dominant selective pressure. Chan *et al.* (2016) reported that most *Umbo-nium* shells from Singapore are light in colour and proposed a thermoregulatory advantage for lighter morphs; the present dataset, which lacks direct thermal measurements or performance trials (adaptive divergence: evidence & limitations part), cannot test this mechanism directly, and the distributional heterogeneity among pale morphs recorded here indicates that thermoregulation is, at best, an incomplete explanation for their spatial distribution in the Sundarbans.

The contribution of developmental plasticity and pseudo-polymorphism remains, as Grüneberg (1980) acknowledged, difficult to disentangle from true genetic polymorphism without breeding experiments or molecular data. The high proportion of variable characters ($H \geq 0.70$; 16 of 25 = 64%) is entirely consistent with Grüneberg's (1980) finding that most *Umbo-nium* variants form a continuous phenotypic sequence without sharp discontinuities; a condition wherein a single genotype may be moulded into multiple phenotypes by environmental or developmental influences. Nevertheless, the extraordinarily high eta-squared values for morphometric parameters ($\eta^2 = 0.435\text{--}0.975$) and the discrete, non-overlapping morph groups in LDA ordination (capturing 94.8% of discrimination variance on LD1 + LD2) indicate the probability, that colour-pattern morph groups are morphometrically real, internally consistent, statistically distinguishable units, and not arbitrary divisions

along a continuous phenotypic continuum. As discussed in adaptive divergence: evidence & limitations part and shell morphometric differentiation among colour-defined morphs part, however, this morphometric distinctiveness, like the entropy-based character patterns discussed in section 4.4, cannot on its own establish that the underlying differences are under genetic, as opposed to environmental or developmental, control; nor does it imply that morphometric and colour-pattern variation co-vary along the same axes, since unsupervised clustering of morphometric data alone does not reproduce the colour-based groupings.

Rare morphs, conservation significance and monitoring

The two rarest morphs in the dataset; Morph 4 ($n = 186$; 4.6% of total) and Morph 9 ($n = 166$; 4.1%) warrant specific conservation attention. Both exhibit narrow island-specific distributions: P4 is largely restricted to Sagar Island, and P9 is concentrated at Sagar and Bakkhali, with complete absence from G Plot. Against the backdrop of the SBR's deteriorating environmental trajectory; rising land surface temperatures ($\sim 0.5^\circ\text{C}$ per decade; Sahana *et al.* 2016), increasing salinity intrusion, mangrove vegetation shifts towards salt-tolerant assemblages (Samanta *et al.* 2021), and projected sea-level rise; morphs with restricted geographic ranges are inherently more vulnerable to local extinction than cosmopolitan forms such as P7. Morph 4's status as the most morphologically isolated taxon in the dataset (mean Jaccard similarity = 0.216) compounds this concern: its unique combination of pink ground colour, uniform stripes, translucent pattern expression, and grey apical colouration would probably be eliminated from the Sundarban gene pool upon local extinction, reducing population-level phenotypic resilience; although that needs further proof from other Islands of Sundarban.

The use of colour morph frequencies as biodiversity indicators has some precedent. Sivadas *et al.* (2012) noted that *U. vestiarium*'s habitat selectivity makes it highly sensitive to habitat disturbance, commercial shell collection (Appukuttan & Ramadoss 2000), and fishing pressure. Hossain and Maxwell (2022) highlighted further vulnerabilities from land-use change, increased turbidity, and over-exploitation. Sandy intertidal habitats globally face mounting threats from sea-level rise, increased storm intensity, coastal armourisation, and direct human disturbance (Defeo *et al.* 2009), within the SBR specifically, sea-level rise is predicted to narrow the intertidal zone by 20–70% depending on coastal geomorphology (Sivadas *et al.* 2012). The morph frequency baselines and island-specific guild assignments established here provide an empirically grounded reference against which future monitoring

surveys can detect compositional change. A significant reduction in morph richness, a collapse in Pielou's evenness, or a shift in group dominance towards a P7 monoculture could serve as sensitive early-warning indicators of intertidal habitat degradation.

Synthesis: the Sundarban tidal mosaic as a polymorphism engine – a hypothesis

he overarching finding of this study is that *U. vestiarium* populations in the Sundarban Biosphere Reserve harbour a structured, multi-dimensional phenotypic polymorphism that cannot be reduced to a simple colour-based description. The eleven morphs documented here are not merely aesthetic variants; they differ in colour pattern, shell architecture, and island-level abundance, although, as shown in hierarchical clustering of mean morphometrics and synthesis: discordance between colour-pattern and morphometric classifications part, these dimensions of variation are not organised along the same group boundaries and should be regarded as partly independent rather than co-varying as a single phenotypic module. The world's largest mangrove delta, with its extraordinary spatial heterogeneity of substrates, salinities, wave regimes, and insolation environments (Hait & Behling 2019; Sahana *et al.* 2016, 2022; Samanta *et al.* 2021), may nonetheless function as a landscape that generates and maintains multiple, partly independent axes of intraspecific phenotypic diversity, through spatially variable selective pressures acting differently on colour and on shell shape. This interpretation resonates with the observation of Grüneberg (1980), who noted that it would be difficult to imagine selective advantages of colour pattern in *Umbonium*, which lives in the sand and seems to have no terrestrial predators, yet whose populations nonetheless harbour remarkable phenotypic diversity across localities. The Sundarban data suggest that the explanation may lie not in a single, integrated adaptive syndrome but in the complexity and heterogeneity of the physical and biological environment acting on different aspects of the phenotype in different ways; a hypothesis that, as outlined in adaptive divergence: evidence & limitations part, requires molecular genetic and experimental testing before it can be confirmed, and that, as environmental change accelerates in the Sundarbans, may become progressively harder to test as habitat heterogeneity itself is eroded.

Conclusion

This study provides the first formally structured, quantitatively grounded characterisation of *Umbonium vestiarium* polymorphism from the Sundarban Biosphere Reserve, combining morphological, morphometric, and distributional analyses across six intertidal islands. Eleven reproducible colour morphs (P1–P11)

were designated using a 25-character binary matrix (inter-observer agreement > 97%), yielding a mean Jaccard similarity of 0.272 ± 0.113 ; establishing these as genuinely differentiated phenotypic entities rather than superficial chromatic variants. UPGMA clustering and PCoA ordination consistently resolved three coherent morph groups: a brown-banded planispiral high-intertidal assemblage (Group I), a translucent hypo-pigmented group including the cosmopolitan dominant P7 (Group II), and a pink-banded turbinate sheltered-intertidal assemblage (Group III).

Ward hierarchical clustering of six shell morphometric parameters resolved a three-group topology that does not correspond to the colour-pattern classification: only P4 and P7 (colour Group II) remain grouped together across both analyses, colour Groups I and III are each split across different morphometric clusters, and P11 is morphometrically isolated as a singleton, the most divergent morph in the dataset (hierarchical clustering of mean morphometrics part). Morph identity nonetheless explains up to 97.5% of total variance in spire height ($\eta^2 = 0.435\text{--}0.975$), confirming that morphometric variation among morphs is real, substantial and internally consistent; it is simply structured differently from colour-pattern variation, indicating that these two dimensions of the phenotype are substantially decoupled rather than co-varying as a single, integrated module. This discordance means the present data cannot be used to argue for, or against, neutral drift and we no longer regard congruence between colour-based and morphometric classification as evidence for phenotypic co-divergence or for a shared selective or genetic basis. Because the present dataset is also correlational and cross-sectional, comprising a single temporal sample with no accompanying genetic or experimental fitness data, it cannot independently distinguish adaptive divergence in either colour pattern or shell morphometry from alternative explanations such as genetic drift in a low-dispersal system, historical or founder effects among islands, or non-adaptive developmental plasticity. We therefore propose, as a hypothesis for future testing rather than an established conclusion, that colour pattern and shell morphometry in this population may respond to partly independent selective, developmental, or genetic influences within the SBR's heterogeneous tidal landscape. Confirming this will require the molecular genetics and experimental approaches outlined in adaptive divergence: evidence & limitations part.

From a conservation standpoint, morph 4 ($n = 186$; mean Jaccard $J = 0.216$; the most morphologically isolated morph) and morph 9 ($n = 166$; restricted to two islands) are identified as priority taxa warranting active monitoring within this UNESCO World Heritage ecosystem. The morph frequency and abundance baselines established here provide the first

quantitative reference for tracking phenotypic biodiversity responses to the SBR's accelerating environmental changes, including rising temperatures, salinity intrusion and projected intertidal habitat contraction. Future priorities include molecular genetic analysis, pigment biochemistry, reciprocal transplant experiments and longitudinal seasonal monitoring to mechanistically validate the mosaic adaptive divergence hypotheses proposed in this study.

Acknowledgements

The authors thank the Director, Zoological Survey of India, Kolkata and Ministry of Environment, Forest and Climate Change, Government of India, for providing facilities and financial support. Authors also thank the Principal Chief Conservator of Forest (West Bengal), Chief Conservator of Forest and Divisional Forest Officer (Sundarban Tiger Reserve and Sundarban Biosphere Reserve) for their permission and cooperation for this study.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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